Chapter 3

Isomorphic Graphs

3.1 The Definition of Isomorphism

Recall that two graphs G and H are equal if $V(G) = V(H)$ and $E(G) = E(H)$. We have called two graphs G and H "isomorphic" if they have the same structure and have written $G \cong H$ to indicate this. That is, $G \cong H$ if the vertices of G and H can be labeled (or relabeled) to produce two equal graphs. We now make all of this more precise.

Suppose that you are asked to give an example of three graphs having order 5 and size 5. Would the three graphs H_1, H_2, and H_3 given in Figure 3.1 be an acceptable answer to this question?

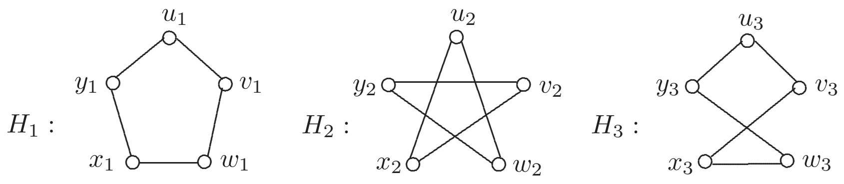


Figure 3.1: Graph(s) of order 5 and size 5

By repositioning the vertices of H_2, we have redrawn H_2 as in Figure 3.2. Similarly, we can redraw H_3 as in Figure 3.2.

It should now be clear that the graphs of Figure 3.1 differ only in the way they are labeled and in the way they are drawn, that is, they certainly have the same structure. In a certain sense then, we have only given an example of one graph of order 5 and size 5. That is, the graphs of Figure 3.1 are simply disguised forms of the same graph, namely, the 5-cycle C_5. For example, the redrawing of H_2 shown in Figure 3.2 suggests that (1) u_2 in H_2 is playing the role of u_1 in H_1, (2) w_2 is playing the role of v_1, (3) y_2 is playing the role of w_1, (4) v_2 is playing

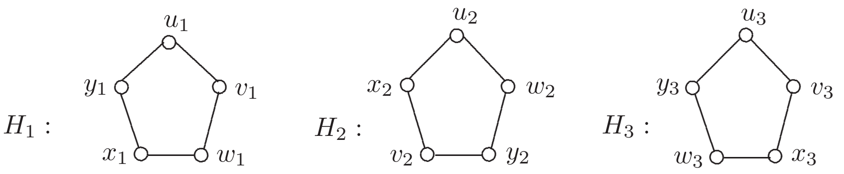


Figure 3.2: A graph of order 5 and size 5

the role of x_1, and (5) x_2 is playing the role of y_1. The manner in which the vertices of H_2 correspond to the vertices of H_1 is not unique, however. Two other drawings of H_2 shown in Figure 3.3 suggest that there are other correspondences between the vertices of H_1 and the vertices of H_2. Indeed, there are several such correspondences.

Figure 3.3: Other drawings of the graph H_2

As we have said, when two graphs differ only in the way they're drawn and/or labeled, then they are said to be isomorphic. Formally, two (labeled) graphs G_1 and G_2 are **isomorphic** (have the same structure) if there exists a one-to-one correspondence ϕ from $V(G_1)$ to $V(G_2)$ such that $u_1v_1 \in E(G_1)$ if and only if $\phi(u_1)\phi(v_1) \in E(G_2)$. In this case, ϕ is called an **isomorphism** from G_1 to G_2. Thus, if G_1 and G_2 are isomorphic graphs, then we say that G_1 **is isomorphic to** G_2 and we write $G_1 \cong G_2$. If G_1 and G_2 are unlabeled, then they are isomorphic if, under any labeling of their vertices, they are isomorphic as labeled graphs. If two graphs G and H are not isomorphic, then they are called **nonisomorphic graphs** and we write $G \ncong H$.

As is implied by the redrawing of the graph H_2 in Figure 3.2, the function $\phi : V(H_1) \to V(H_2)$ defined by

$$\phi(u_1) = u_2,\ \phi(v_1) = w_2,\ \phi(w_1) = y_2,\ \phi(x_1) = v_2,\ \phi(y_1) = x_2$$

is an isomorphism and so $H_1 \cong H_2$. Intuitively then, two graphs are isomorphic if it is possible to redraw one of them so that the two diagrams are the same. This highly informal interpretation of isomorphism, although often suitable, is not satisfactory in all cases, and so it may be necessary to rely on the formal definition. In this context then, we consider two graphs to be the "same" graph if they are isomorphic and to be "different" if they are not isomorphic. From this point of view, there is only one graph of order 1, two graphs of order 2, and four graphs of order 3. There are eleven (nonisomorphic) graphs of order 4 and these are shown in Figure 3.4.

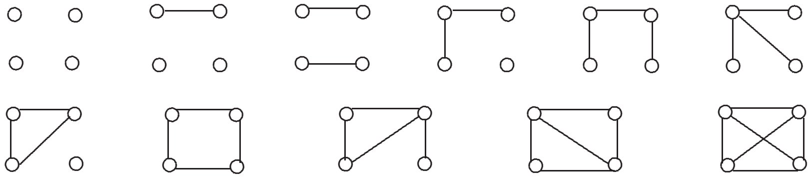


Figure 3.4: The eleven graphs of order 4

Let's look at the definition of isomorphism more closely. First, in order for two graphs G_1 and G_2 to be isomorphic, there must be a one-to-one correspondence from the vertex set of G_1 to the vertex set of G_2. This means that it must be possible to pair off the vertices of G_1 with the vertices of G_2. Therefore, $|V(G_1)| = |V(G_2)|$ and so G_1 and G_2 have the same order. It is certainly not surprising that we would want two graphs to be of the same order if we want to consider them to be the same graph.

Continuing to analyze the definition of isomorphic graphs, we see that not only must there be a one-to-one correspondence from $V(G_1)$ to $V(G_2)$ but two vertices u_1 and v_1 of G_1 are adjacent in G_1 if and only if the corresponding vertices $\phi(u_1)$ and $\phi(u_2)$ are adjacent in G_2. So adjacent vertices in G_1 are mapped to adjacent vertices in G_2, while nonadjacent vertices in G_1 are mapped to nonadjacent vertices in G_2. This implies that for G_1 and G_2 to be isomorphic, they must have the same size – again not a particularly surprising piece of information.

Hence if two graphs are isomorphic, then they must have the same order and the same size. The contrapositive of this statement says that if two graphs have different orders or different sizes, then they are not isomorphic. For example, even though the graphs F' and F'' of Figure 3.5 have the same size 6, they are not isomorphic because their orders are different. Also, the graphs H' and H'' of Figure 3.5 have the same order 6 but cannot be isomorphic since their sizes are different.

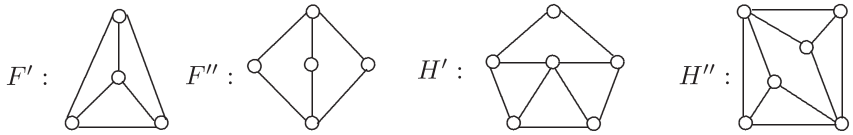


Figure 3.5: Nonisomorphic graphs

On the other hand, if two graphs have the same order and the same size, then there is no guarantee that the graphs are isomorphic. For example, the graphs G_1 and G_2 of Figure 3.6 have order 6 and size 6, yet they are not isomorphic. In order to see this, assume, to the contrary, that they are isomorphic. Then there exists an isomorphism $\phi : V(G_1) \to V(G_2)$. Hence there are three vertices of G_1 that map into u_2, v_2, z_2 of G_2. Since u_2, v_2, and z_2 are pairwise adjacent and form a triangle, so too are the vertices of G_1 that map into these three vertices of G_2. However, G_1 doesn't contain a triangle, and a contradiction is produced.

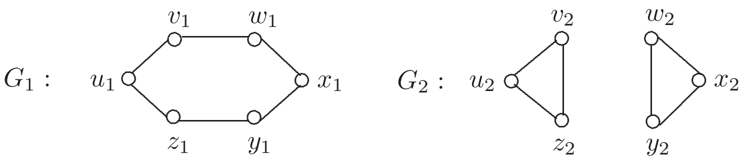


Figure 3.6: Two nonisomorphic graphs

Let's revisit the definition of isomorphism yet again. Two graphs G_1 and G_2 are isomorphic if there exists a one-to-one correspondence ϕ from $V(G_1)$ to $V(G_2)$ such that every two adjacent vertices of G_1 are mapped to adjacent vertices of G_2 and every two nonadjacent vertices of G_1 are mapped to nonadjacent vertices of G_2. Recall that a function ϕ with these properties is an isomorphism. However, since $V(\overline{G}_1) = V(G_1)$ and $V(\overline{G}_2) = V(G_2)$, the same function $\phi : V(\overline{G}_1) \to V(\overline{G}_2)$ also maps adjacent vertices of $\overline{G}_1$ to adjacent vertices of $\overline{G}_2$ and nonadjacent vertices of $\overline{G}_1$ to nonadjacent vertices of $\overline{G}_2$. This observation provides us with the following theorem.

Theorem 3.1 *Two graphs G_1 and G_2 are isomorphic if and only if their complements $\overline{G}_1$ and $\overline{G}_2$ are isomorphic.*

Let's consider the two graphs H_1 and H_2 shown in Figure 3.7. Both graphs have order 6 and size 9; so H_1 and H_2 might be isomorphic, but we don't know this for sure. Since $\overline{H}_1 = G_1$ and $\overline{H}_2 = G_2$ (where G_1 and G_2 are the graphs shown in Figure 3.6) and G_1 and G_2 are not isomorphic, it follows by Theorem 3.1 that H_1 and H_2 are also not isomorphic. It is possible to see that H_1 and H_2 are not isomorphic without the aid of Theorem 3.1, however. Assume, to the contrary, that H_1 and H_2 are isomorphic. Then there exists an isomorphism $\phi : V(H_1) \to V(H_2)$. The vertices v_1, x_1, and z_1 are mutually adjacent in H_1 and form a triangle, and so $\phi(v_1), \phi(x_1)$, and $\phi(z_1)$ form a triangle in H_2. However, H_2 contains no triangle, and a contradiction is produced.

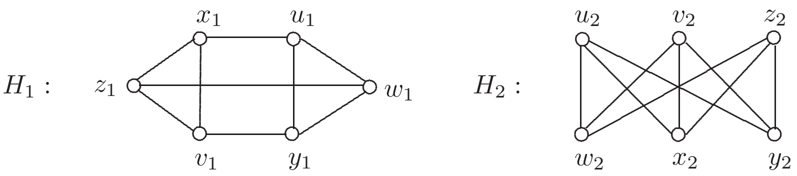


Figure 3.7: Two graphs H_1 and H_2

A graph and its complement may, in fact, be isomorphic to each other. A graph G is **self-complementary** if $G \cong \overline{G}$. Of course, this can only occur if G and $\overline{G}$ have the same size, namely $\frac{1}{2}\binom{n}{2} = \frac{n(n-1)}{4}$. For $\frac{n(n-1)}{4}$ to be an integer, either $4 \mid n$ or $4 \mid (n-1)$, that is, either $n \equiv 0 \pmod 4$ or $n \equiv 1 \pmod 4$. Figure 3.8 shows four self-complementary graphs.

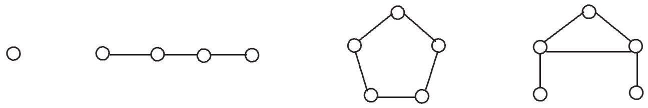


Figure 3.8: Self-complementary graphs

Not only are the orders the same and the sizes the same of two isomorphic graphs, so too are the degrees of their vertices.

Theorem 3.2 *If G and H are isomorphic graphs, then the degrees of the vertices of G are the same as the degrees of the vertices of H.*

Proof. [direct proof] Since G and H are isomorphic, there is an isomorphism $\phi : V(G) \to V(H)$. Let u be a vertex of G and suppose that $\phi(u) = v$, where v therefore belongs to H. We show that the degree of v in H is the same as the degree of u in G. Suppose first that $\deg_G u = 0$. Then no vertex of G is adjacent to u. Let y be any vertex of H that is different from v. Then there is a vertex x in G such that $\phi(x) = y$. Since x and u are not adjacent in G, the vertices y and v are not adjacent in H. So no vertex of H is adjacent to v and $\deg_H v = 0$.

Suppose next that $\deg_G u = k \geq 1$. Let $N(u) = \{x_1, x_2, \ldots, x_k\}$ be the set of vertices adjacent to u in G, where $\phi(x_i) = y_i$ for $1 \leq i \leq k$. Since u and x_i are adjacent for $1 \leq i \leq k$, the vertices v and y_i are adjacent for $1 \leq i \leq k$. If y is a vertex of H such that $y \notin \{v\} \cup \{y_1, y_2, \ldots, y_k\}$, then there is a vertex x in G such that $\phi(x) = y$, where $x \notin \{u\} \cup N(u)$. Since x and u are not adjacent in G, the vertices v and y are not adjacent in H. So $\deg_H v = k$. ■

Theorem 3.2 not only tells us that two isomorphic graphs G and H have the same degree sequence but the proof of this theorem also says that if ϕ is an isomorphism from $V(G)$ to $V(H)$ and u is a vertex of G, then $\deg_G u = \deg_H \phi(u)$, that is, under an isomorphism a vertex can only map into a vertex having the same degree.

So now we know that if G and H are isomorphic graphs, then their orders are the same, their sizes are the same, and the degrees of their vertices are the same. On the other hand, if the degrees of their vertices are the same, then their orders must be the same and their sizes must also be the same. As with the order and size, two graphs having the same degree sequences is only a necessary condition, not a sufficient condition, for two graphs to be isomorphic. For example, the degree sequences of the nonisomorphic graphs G_1 and G_2 of Figure 3.6 are both $2, 2, 2, 2, 2, 2$; while the degree sequences of the nonisomorphic graphs of Figure 3.7 are both $3, 3, 3, 3, 3, 3$.

Therefore, the challenge for determining whether two graphs are isomorphic is when the two graphs have the same degree sequence. Let's consider some examples of this.

Example 3.3 *Determine whether the graphs F_1 and F_2 of Figure 3.9 are isomorphic.*

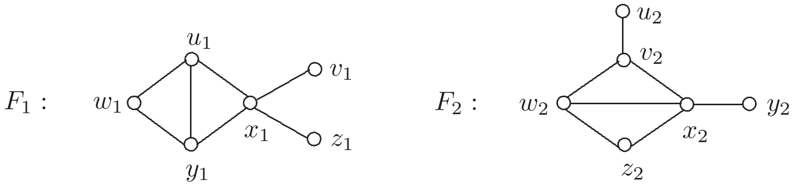


Figure 3.9: The two graphs in Example 3.3

Solution. Since both F_1 and F_2 have the degree sequence $4, 3, 3, 2, 1, 1$, they may be isomorphic. But they are not. Assume, to the contrary, that $F_1 \cong F_2$. Then there exists an isomorphism $\phi : V(F_1) \to V(F_2)$. The vertex x_1 is the only vertex of F_1 having degree 4. Thus $\phi(x_1)$ has degree 4 in F_2. Since x_2 is the only vertex of F_2 having degree 4, it follows that $\phi(x_1) = x_2$. Since both v_1 and z_1 are adjacent to x_1 in F_1, both $\phi(v_1)$ and $\phi(z_1)$ are adjacent to $\phi(x_1) = x_2$ in F_2. Because $\deg_{F_1} v_1 = \deg_{F_1} z_1 = 1$, it is also the case that $\deg_{F_2} \phi(v_1) = \deg_{F_2} \phi(z_1) = 1$. But this says that x_2 is adjacent to two end-vertices in F_2, which is not the case. This is a contradiction. $\diamondsuit$

The argument just given to show that the graphs F_1 and F_2 of Figure 3.9 are not isomorphic can be simplified. The vertex of degree 4 in F_1 is adjacent to two end-vertices; while the vertex of degree 4 in F_2 is not. Therefore, F_1 and F_2 are not isomorphic.

Indeed, let G_1 and G_2 be two graphs. We might as well assume that G_1 and G_2 have the same degree sequence; otherwise, we know immediately that $G_1 \ncong G_2$. If G_1 has some property that doesn't depend on how G_1 is drawn or how G_1 is labeled and G_2 does not have this property, then $G_1 \ncong G_2$. For example, if G_1 contains two vertices of degree 3 that are mutually adjacent to a vertex of degree 2 and G_2 does not, then $G_1 \ncong G_2$. If G_1 contains two triangles that have a common vertex and G_2 doesn't, then $G_1 \ncong G_2$. If G_1 contains eight triangles and G_2 contains only seven triangles, then $G_1 \ncong G_2$. This last statement brings up an important point, however. If the explanation that was given as to why two graphs are not isomorphic is that one has eight triangles and the other has seven triangles, then this is probably not convincing since it may not be clear that all triangles in both graphs have been accounted for. It would be preferable to locate a property that is easier to justify (assuming that the two graphs are in fact not isomorphic).

Example 3.4 *Determine whether the graphs H_1 and H_2 of Figure 3.10 are isomorphic.*

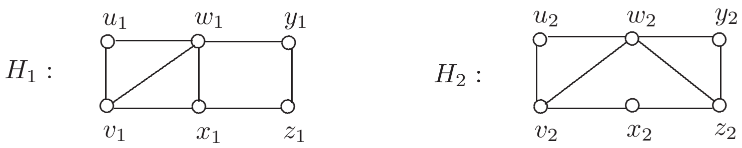


Figure 3.10: The two graphs in Example 3.4

Solution. First, observe that H_1 and H_2 have the degree sequence $4, 3, 3, 2, 2, 2$. Hence further consideration is needed. Because these two graphs do not "appear" to be isomorphic, we seek some structural difference. Observe that H_1 contains two adjacent vertices of degree 2 (namely y_1 and z_1), while H_2 does not. Thus $H_1 \not\cong H_2$. $\diamondsuit$

What we are observing is that if G and H are isomorphic graphs, $\phi : V(G) \to V(H)$ is an isomorphism, and the vertex u of G is mapped by ϕ to the vertex v of H, then any property that u has in G must be a property that v has in H provided this property doesn't depend on how the graphs are drawn or labeled. More generally, any structural property of G must also be possessed by H. For example,

(1) if G contains a k-cycle for some integer $k \geq 3$, so does H, and

(2) if G contains a $u - v$ path of length k, then H contains a $\phi(u) - \phi(v)$ path of length k.

These remarks give the following theorem.

Theorem 3.5 *Let G and H be isomorphic graphs. Then*

(a) G is bipartite if and only if H is bipartite, and

(b) G is connected if and only if H is connected.

As expected, two digraphs D_1 and D_2 are **isomorphic** if there exists a one-to-one correspondence $\phi : V(D_1) \to V(D_2)$ such that $(u_1, v_1) \in E(D_1)$ if and only if $(\phi(u_1), \phi(v_1)) \in E(D_2)$. Digraphs will be studied in detail in Chapter 7.

Exercises for Section 3.1

3.1 Give an example of three different (nonisomorphic) graphs of order 5 and size 5.

3.2 Give an example of three graphs of the same order, same size, and same degree sequence such that no two of these graphs are isomorphic.

3.3 For each of the pairs G_1, G_2 of graphs in Figures 3.11(a) and 3.11(b), determine (with careful explanation) whether G_1 and G_2 are isomorphic.

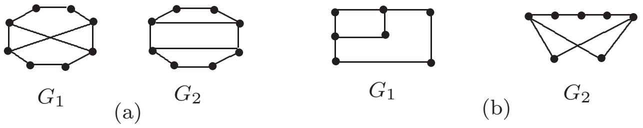


Figure 3.11: The graphs in Exercise 3.3

3.4 Which pairs of graphs in Figure 3.12 are isomorphic? Explain your answer.

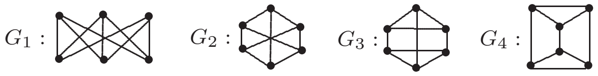


Figure 3.12: The graphs in Exercise 3.4

3.5 Let G_1 and G_2 be two graphs with $V(G_1) = \{u_1, v_1, w_1, x_1, y_1, z_1\}$ and $V(G_2) = \{u_2, v_2, w_2, x_2, y_2, z_2\}$. If v_1 has degree 3 and is adjacent to a vertex of degree 2, while v_2 has degree 3 and is not adjacent to a vertex of degree 2, can we conclude that $G_1 \not\cong G_2$? Explain your answer.

3.6 Let G_1 and G_2 be two graphs having the same degree sequence. If G_1 contains a vertex of degree 2 that is adjacent to a vertex of degree 3 and a vertex of degree 4, while G_2 contains a vertex of degree 2 that is adjacent to two vertices of degree 3, can we conclude that $G_1 \not\cong G_2$? Explain your answer.

3.7 Is the solution of the following problem correct?

Problem: Determine whether the graphs G_1 and G_2 of Figure 3.13 are isomorphic.

Figure 3.13: The two graphs in Exercise 3.7

Solution. The graph G_1 has a 5-cycle C. Two vertices of C are connected by a path of length 2, lying inside C. The graph G_2 does not contain such a 5-cycle, however. Therefore, $G_1 \not\cong G_2$. $\diamondsuit$

3.8 Which pairs of graphs in Figure 3.14 are isomorphic? Explain your answer.

3.9 Determine whether the graphs G_1 and G_2 of Figure 3.15 are isomorphic. Explain your answer.

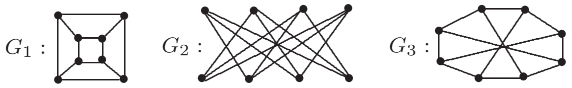


Figure 3.14: Graphs in Exercise 3.8

Figure 3.15: Two graphs in Exercise 3.9

3.10 Does there exist a disconnected self-complementary graph?

3.11 Let G be a self-complementary graph of order $n = 4k$, where $k \geq 1$. Let $U = \{v : \deg v \leq n/2\}$ and $W = \{v : \deg v \geq n/2\}$. Prove that if $|U| = |W|$, then G contains no vertex v such that $\deg v = n/2$.

3.12 Let G and H be two self-complementary graphs with disjoint vertex sets, where H has even order n. Let F be the graph obtained from $G \cup H$ by joining each vertex of G to every vertex of degree less than $n/2$ in H. Show that F is self-complementary.

3.13 Suppose that there exist two connected graphs G and H and a one-to-one function ϕ from $V(G)$ onto $V(H)$ such that $d_G(u, v) = d_H(\phi(u), \phi(v))$ for every two vertices u and v of G. Prove or disprove: G and H are isomorphic.

3.14 Prove or disprove: Let G and H be two connected graphs of order n, where $V(G) = \{v_1, v_2, \ldots, v_n\}$. If there exists a one-to-one correspondence $\phi : V(G) \to V(H)$ such that $d_G(v_i, v_{i+1}) = d_H(\phi(v_i), \phi(v_{i+1}))$ for all i $(1 \leq i \leq n-1)$, then $G \cong H$.

3.15 Prove or disprove: Let G and H be two connected graphs. If there exists a one-to-one correspondence $\phi : V(G) \to V(H)$ and two distinct vertices $u, v \in V(G)$ such that $d_G(u, v) \neq d_H(\phi(u), \phi(v))$, then $G \ncong H$.

3.2 Isomorphism as a Relation

Let us state once again that a graph G_1 is isomorphic to a graph G_2 if there exists an isomorphism $\phi : V(G_1) \to V(G_2)$. Isomorphism therefore produces a relation on any set of graphs, namely, a graph G_1 is related to a graph G_2

if G_1 is isomorphic to G_2. This relation is an equivalence relation. This says that isomorphism is reflexive (every graph is isomorphic to itself), isomorphism is symmetric (if G_1 is isomorphic to G_2, then G_2 is isomorphic to G_1), and isomorphism is transitive (if G_1 is isomorphic to G_2 and G_2 is isomorphic to G_3, then G_1 is isomorphic to G_3).

The proof that isomorphism is an equivalence relation relies on three fundamental properties of bijective functions (functions that are one-to-one and onto): (1) every identity function is bijective; (2) every bijective function has an inverse that is also bijective; (3) the composition of two bijective functions is bijective. (See Appendix 2 for a review of these terms and facts.)

Theorem 3.6 *Isomorphism is an equivalence relation on the set of all graphs.*

Proof. [direct proof] First, we show that isomorphism is reflexive, that is, every graph is isomorphic to itself. Let G be a graph and consider the identity function $\epsilon : V(G) \to V(G)$ defined by $\epsilon(v) = v$ for each vertex v of G. Thus ϵ is bijective. Certainly, two vertices u and v of G are adjacent if and only if $\epsilon(u) = u$ and $\epsilon(v) = v$ are adjacent. Therefore, ϵ is an isomorphism and so G is isomorphic to G.

Next, we show that isomorphism is symmetric. Let G_1 and G_2 be graphs, and assume that G_1 is isomorphic to G_2. Therefore, there exists an isomorphism $\phi : V(G_1) \to V(G_2)$. Since ϕ is a bijective function, its inverse $\phi^{-1} : V(G_2) \to V(G_1)$ exists and is a bijective function. Let u_2 and v_2 be any two vertices in G_2 and suppose that $\phi^{-1}(u_2) = u_1$ and $\phi^{-1}(v_2) = v_1$. Hence $\phi(u_1) = u_2$ and $\phi(v_1) = v_2$. If u_2 and v_2 are adjacent vertices in G_2, then u_1 and v_1 are adjacent vertices in G_1 since ϕ is an isomorphism. On the other hand, if u_2 and v_2 are not adjacent, then u_1 and v_1 are not adjacent. Therefore, u_2 and v_2 are adjacent in G_2 if and only if $\phi^{-1}(u_2)$ and $\phi^{-1}(v_2)$ are adjacent in G_1. Hence ϕ^{-1} is an isomorphism and so G_2 is isomorphic to G_1.

Finally, we show that isomorphism is transitive. For graphs G_1, G_2, and G_3, assume that G_1 is isomorphic to G_2 and G_2 is isomorphic to G_3. Hence there exist isomorphisms $\alpha : V(G_1) \to V(G_2)$ and $\beta : V(G_2) \to V(G_3)$. Consider the composition $\beta \circ \alpha : V(G_1) \to V(G_3)$. Since α and β are bijective, so is $\beta \circ \alpha$. Since α is an isomorphism, vertices u_1 and v_1 of G_1 are adjacent if and only if $\alpha(u_1)$ and $\alpha(v_1)$ are adjacent in G_2. Because β is an isomorphism, $\alpha(u_1)$ and $\alpha(v_1)$ are adjacent in G_2 if and only if $\beta(\alpha(u_1))$ and $\beta(\alpha(v_1))$ are adjacent in G_3. Therefore, u_1 and v_1 are adjacent in G_1 if and only if $(\beta \circ \alpha)(u_1)$ and $(\beta \circ \alpha)(v_1)$ are adjacent in G_3 and so $\beta \circ \alpha$ is an isomorphism. Hence G_1 is isomorphic to G_3. ∎

One of the major consequences of knowing that isomorphism is an equivalence relation on a set of graphs is that this produces a partition of this set into equivalence classes (subsets), which we called **isomorphism classes** here. Every two graphs in the same isomorphism class are isomorphic and every two graphs in different isomorphism classes are not isomorphic.

Suppose that we are asked for all graphs with degree sequence s: 2, 2, 2, 2, 2, 2, 2, 2, 2. What we are clearly seeking here are nonisomorphic graphs. The answer to this question is given in Figure 3.16. (There are four such graphs!)

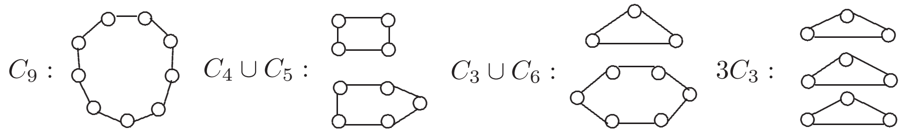


Figure 3.16: All graphs with degree sequence $s : 2, 2, 2, 2, 2, 2, 2, 2, 2$

Consider the graphs H and G of Figure 3.17. Is H a subgraph of G? This may seem confusing since we defined H to be a subgraph of G, written $H \subseteq G$, if $V(H) \subseteq V(G)$ and $E(H) \subseteq E(G)$. This, in fact, *is* the definition if the graphs H and G are labeled (that is, if the vertex sets of H and G have been specified). The graphs H and G of Figure 3.17 are not labeled, however. For unlabeled graphs H and G, we say that H is a **subgraph** of G if for any labeling of the vertices of H and G, the graph H is isomorphic to a subgraph of G. Consequently, for the graphs H and G of Figure 3.17, H *is* a subgraph of G.

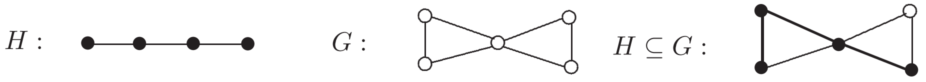


Figure 3.17: A subgraph H of a graph G

Exercises for Section 3.2

3.16 How many (nonisomorphic) graphs have the degree sequence s: 6, 6, 6, 6, 6, 6, 6, 6, 6?

3.17 Consider the (unlabeled) graphs H_1, H_2, H_3, and G of Figure 3.18.

(a) Is H_1 a subgraph of G?

(b) Is H_2 a subgraph of G?

(c) Is H_3 a subgraph of G?

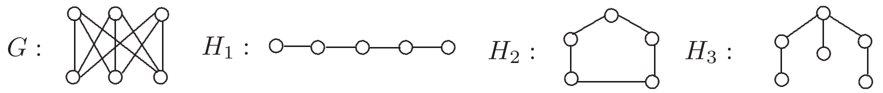


Figure 3.18: Graphs in Exercise 3.17

3.18 Does there exist a graph with exactly three components, exactly two of which are not isomorphic?

3.19 We are given a collection of n graphs $G_1, G_2, \ldots, G_n$, some pairs of which are isomorphic and some pairs of which are not. Show that there are an even number of these graphs that are isomorphic to an odd number of graphs. [Hint: Construct a graph G with $V(G) = \{v_1, v_2, \ldots, v_n\}$, where $v_iv_j \in E(G)$ if and only if G_i is isomorphic to G_j.]

3.3 Excursion: Graphs and Groups

While it may be quite difficult to determine that two isomorphic graphs G_1 and G_2 are in fact isomorphic if G_1 and G_2 are drawn differently or labeled differently, there is no difficulty in showing that G_1 and G_2 are isomorphic if they are drawn and labeled identically. In this case, we then have a single graph, say G, and surely the identity function $\epsilon : V(G) \to V(G)$, where $\epsilon(v) = v$ for all $v \in V(G)$, is an isomorphism. Consequently, for the graph H of Figure 3.19, the function $\alpha_1 : V(H) \to V(H)$ defined by $\alpha_1(v) = v$ for every vertex v of H is an isomorphism.

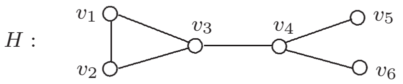


Figure 3.19: A graph H

There are other isomorphisms from the graph H of Figure 3.19 to itself, however. For example, the function $\alpha_2 : V(H) \to V(H)$ defined by

$$\alpha_2(v) = \begin{cases} v_2 & \text{if } v = v_1 \\ v_1 & \text{if } v = v_2 \\ v & \text{if } v \neq v_1, v_2 \end{cases}$$

is an isomorphism as well. There are two other isomorphisms from the graph H to itself, namely α_3 and α_4, defined by

$$\alpha_3(v) = \begin{cases} v_6 & \text{if } v = v_5 \\ v_5 & \text{if } v = v_6 \\ v & \text{if } v \neq v_5, v_6 \end{cases}$$

and

$$\alpha_4(v) = \begin{cases} v_2 & \text{if } v = v_1 \\ v_1 & \text{if } v = v_2 \\ v_6 & \text{if } v = v_5 \\ v_5 & \text{if } v = v_6 \\ v & \text{if } v = v_3, v_4. \end{cases}$$

An isomorphism from a graph G to itself is called an **automorphism** of G. Since composition is associative, the identity function is an automorphism, the inverse of an automorphism is an automorphism, and the composition of two automorphisms is an automorphism, it follows that the set of all automorphisms of a graph G forms a group under the operation of composition. This group is denoted by $\mathrm{Aut}(G)$ and is called the **automorphism group** of G. For example, for the graph H of Figure 3.19, $\mathrm{Aut}(H) = \{\alpha_1, \alpha_2, \alpha_3, \alpha_4\}$.

Since every automorphism of a graph is a permutation on $V(G)$, automorphisms can be expressed, more simply, in terms of permutation cycles. (See Appendix 2 for a review of permutations.) For the graph H of Figure 3.19, the elements of $\mathrm{Aut}(H)$ can be expressed as

$$\alpha_1 = \epsilon \text{ (the identity)},\ \alpha_2 = (v_1\ v_2),\ \alpha_3 = (v_5\ v_6),\ \alpha_4 = (v_1\ v_2)(v_5\ v_6).$$

For example, expressing α_4 as the "product" of the permutation cycles $(v_1\ v_2)$ and $(v_5\ v_6)$ means that (1) α_4 maps v_1 into v_2, and v_2 into v_1, (2) α_4 maps v_5 into v_6, and v_6 into v_5, and (3) α_4 fixes all other vertices of H (that is, α_4 maps any other vertex of H into itself).

The group table for the automorphism group of the graph H of Figure 3.19 is shown in Figure 3.20. The reason for the entry α_2 in row 4, column 3 of the group table is because the "product" of α_4 and α_3 is α_2. That is, since

$$\begin{aligned}
(\alpha_4\alpha_3)(v_1) &= \alpha_4(\alpha_3(v_1)) = \alpha_4(v_1) = v_2\\
(\alpha_4\alpha_3)(v_2) &= \alpha_4(\alpha_3(v_2)) = \alpha_4(v_2) = v_1\\
(\alpha_4\alpha_3)(v_3) &= \alpha_4(\alpha_3(v_3)) = \alpha_4(v_3) = v_3\\
(\alpha_4\alpha_3)(v_4) &= \alpha_4(\alpha_3(v_4)) = \alpha_4(v_4) = v_4\\
(\alpha_4\alpha_3)(v_5) &= \alpha_4(\alpha_3(v_5)) = \alpha_4(v_6) = v_5\\
(\alpha_4\alpha_3)(v_6) &= \alpha_4(\alpha_3(v_6)) = \alpha_4(v_5) = v_6,
\end{aligned}$$

it follows that $\alpha_4\alpha_3 = \alpha_2$. However, this can be seen more easily when the automorphisms are expressed in terms of permutation cycles. Since

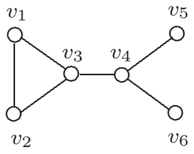
$$\alpha_4\alpha_3 = [(v_1\ v_2)(v_5\ v_6)](v_5\ v_6),$$

we can see, reading from right to left, that (1) v_1 is first mapped into v_2 and then into v_1, (2) v_3 and v_4 are fixed throughout, (3) v_5 is mapped into v_6 by $(v_5\ v_6)$, v_6 is mapped into v_5 by $(v_5\ v_6)$, and v_5 is fixed by $(v_1\ v_2)$, resulting in v_5 being mapped into v_5, and similarly, (4) v_6 is fixed. That is,

$$\alpha_4\alpha_3 = (v_1\ v_2)(v_5\ v_6)(v_5\ v_6) = (v_1\ v_2) = \alpha_2.$$

As another illustration of an automorphism group, consider the graph F of Figure 3.21. The elements of $\mathrm{Aut}(F)$ and the group table are given in that figure as well. For example, the automorphism β_5 maps u to v, maps v to w, and maps w to u, leaving all other vertices fixed. If we write β for β_5, then $\beta^2 = \beta\beta = \beta_6$. Furthermore, if we write α for β_2, then $\beta_3 = \beta_5\beta_2 = \beta\alpha$ and $\beta_4 = \alpha\beta$. Consequently, each of the elements of $\mathrm{Aut}(F)$ can be expressed in

$H:$



$\alpha_1 = \epsilon$

$\alpha_2 = (v_1\ v_2)$

$\alpha_3 = (v_5\ v_6)$

$\alpha_4 = (v_1\ v_2)(v_5\ v_6)$

	α_1	α_2	α_3	α_4
α_1	α_1	α_2	α_3	α_4
α_2	α_2	α_1	α_4	α_3
α_3	α_3	α_4	α_1	α_2
α_4	α_4	α_3	α_2	α_1

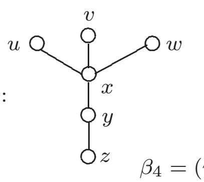
Figure 3.20: The group table for Aut(H)

terms of α and β, namely,

$$\begin{array}{lll} \beta_1 = \alpha^2 = \beta^3, & \beta_2 = \alpha, & \beta_3 = \beta\alpha = \alpha\beta^2, \\ \beta_4 = \alpha\beta = \beta^3\alpha, & \beta_5 = \beta, & \beta_6 = \beta^2. \end{array}$$

Because of this property, α and β are **generators** for the group Aut(F) and $\{\alpha, \beta\}$ is a **generating set** for this group.

$F:$



$\beta_1 = \epsilon$

$\alpha = \beta_2 = (u\ v)$

$\beta_3 = (u\ w)$

$\beta_4 = (v\ w)$

$\beta = \beta_5 = (u\ v\ w)$

$\beta_6 = (u\ w\ v)$

	β_1	β_2	β_3	β_4	β_5	β_6
β_1	β_1	β_2	β_3	β_4	β_5	β_6
β_2	β_2	β_1	β_6	β_5	β_4	β_3
β_3	β_3	β_5	β_1	β_6	β_2	β_4
β_4	β_4	β_6	β_5	β_1	β_3	β_2
β_5	β_5	β_3	β_4	β_2	β_6	β_1
β_6	β_6	β_4	β_2	β_3	β_5	β_5

Figure 3.21: The graph F and the group table for Aut(F)

For a vertex v of a graph G, the set of all vertices into which v can be mapped by some automorphism of G is an **orbit** of G. In fact, if a relation R is defined on $V(G)$ by $x\ R\ y$ if $\alpha(x) = y$ for some $\alpha \in$ Aut(G), then R is an equivalence relation on $V(G)$. The distinct equivalence classes resulting from this relation are the orbits of G. Two vertices u and v are **similar** if they belong to the same orbit. For the graph H of Figure 3.20, there are four orbits, namely, $\{v_1, v_2\}$, $\{v_3\}$, $\{v_4\}$, and $\{v_5, v_6\}$; while for the graph F of Figure 3.21, there are also four orbits: $\{u, v, w\}$, $\{x\}$, $\{y\}$, $\{z\}$. There can be great advantages to knowing the orbits of a graph G. If it is useful to have some structural information about each vertex of G, then it may not be necessary to consider all vertices of G because of the similarity of certain vertices. In such a case, we need only consider one vertex from each orbit as a representative of the orbit. If a graph G of order n has n distinct orbits, then Aut(G) consists of a single automorphism, namely, the identity automorphism. The graph G of Figure 3.22 is one such graph.

$G:$

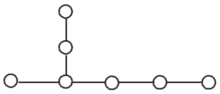


Figure 3.22: A graph of order 7 with seven distinct orbits

On the other hand, if a graph G contains a single orbit, then every two vertices of G are similar and G is called **vertex-transitive**. The graph G_1 of Figure 3.23 is vertex-transitive. Since an automorphism can only map a vertex into a vertex of the same degree, every vertex-transitive graph is regular. The converse is not true, however. For example, the 3-regular graph G_2 of Figure 3.23 is not vertex-transitive since, for example, u belongs to a triangle of G_2 (in fact, two triangles), while w belongs to no triangle of G_2. Therefore, no automorphism of G_2 maps u into w. Among the well-known vertex-transitive graphs are the complete graphs, the cycles, the complete bipartite graphs $K_{s,s}$, and the Petersen graph.

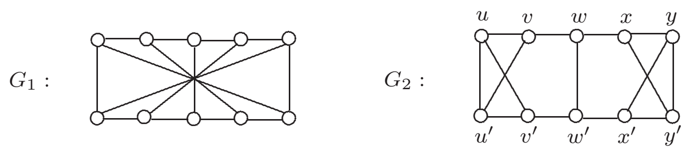


Figure 3.23: A vertex-transitive graph and
a regular graph that is not vertex-transitive

A few fundamental ideas from group theory are useful to review, beginning with the definition of a group itself. A **group** is a nonempty set A (finite or infinite) together with an associative binary operation $\circ$ on A containing an identity element e (necessarily unique) such that $e \circ a = a \circ e = a$ for every element $a \in A$ and having the added property that for every element $a \in A$, there exists an inverse element b (necessarily unique) in A with $b \circ a = a \circ b = e$. Such a group is often denoted by $(A, \circ)$. Because the operation $\circ$ is associative, $(x \circ y) \circ z = x \circ (y \circ z)$ for all $x, y, z \in A$. If $a \circ b = b \circ a$ for all $a, b \in A$, then $(A, \circ)$ is an **abelian group**. The group $\mathrm{Aut}(H)$ for the graph H of Figure 3.20 is abelian, while $\mathrm{Aut}(F)$ is a nonabelian group for the graph F of Figure 3.21. (If the main topic of this text had been group theory, then G would have been used to denote a group. However, graphs have the priority here!)

If $(A, \circ)$ is a group and B is a subset of A such that $(B, \circ)$ is a group, then $(B, \circ)$ is called is a **subgroup** of A. Two groups $(A, \circ)$ and $(B, *)$ are **isomorphic** if there exists a bijective function $\phi : A \to B$ such that $\phi(x \circ y) = \phi(x) * \phi(y)$ for all $x, y \in A$. It is often common to refer to the operation $\circ$ on a group $(A, \circ)$ as multiplication and write $a \circ b$ as ab instead. In this case, we commonly denote the group by A, with the operation understood.

A common type of group to consider is a group of permutations, where the operation is composition. In fact, a well-known theorem of Arthur Cayley states that every group is isomorphic to a group of permutations. As we have mentioned, the automorphism group of a graph is a permutation group. The group of *all* permutations on a set of cardinality n is called the **symmetric group** S_n and its order is $n!$. The automorphism group of a graph G of order n is a group of permutations on the vertex set $V(G)$, that is, the automorphism group of a graph G of order n is a subgroup of S_n. By a theorem of Joseph-Louis Lagrange, the order of $\mathrm{Aut}(G)$ divides $n!$ (the order of S_n).

Since the automorphism group of every graph has finite order, we will only be interested in finite groups. If a group of order n has a single generator, then the group is **cyclic** of order n. Neither $\mathrm{Aut}(H)$ nor $\mathrm{Aut}(F)$ for the graphs H and F of Figures 3.20 and 3.21, respectively, are cyclic. In particular, $\mathrm{Aut}(H)$ is the so-called **Klein four group** (named for Felix Klein), while $\mathrm{Aut}(F)$ is the symmetric group S_3 of order 6. There is another interesting class of groups that appears often in group theory. First, we consider a member of this class.

Example 3.7 *Determine* $\mathrm{Aut}(C_5)$.

Solution. Let $G \cong C_5$, where the vertices of G is labeled as in Figure 3.24.

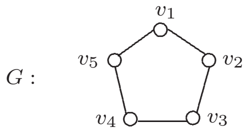


Figure 3.24: The graph G of Example 3.7

One of the automorphisms of G is $\alpha = (v_1\ v_2\ v_3\ v_4\ v_5)$, which can be thought of as a "rotation" of G. Another automorphism of G is $\beta_1 = (v_2\ v_5)(v_3\ v_4)$, which can be thought of as a "reflection" of G. The automorphism group of G consists of the identity ϵ, the four rotations

$$\begin{array}{ll} \alpha = (v_1\ v_2\ v_3\ v_4\ v_5), & \alpha^2 = (v_1\ v_3\ v_5\ v_2\ v_4), \\ \alpha^3 = (v_1\ v_4\ v_2\ v_5\ v_3), & \alpha^4 = (v_1\ v_5\ v_4\ v_3\ v_2) \end{array}$$

and the five reflections

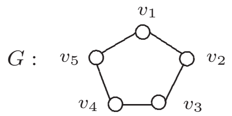
$$\begin{array}{lll} \beta_1 = (v_2\ v_5)(v_3\ v_4), & \beta_2 = (v_1\ v_3)(v_4\ v_5), & \beta_3 = (v_1\ v_5)(v_2\ v_4), \\ \beta_4 = (v_1\ v_2)(v_3\ v_5), & \beta_5 = (v_1\ v_4)(v_2\ v_3). & \end{array}$$

Letting $\beta = \beta_1$, we see in the group table of $\mathrm{Aut}(G)$ shown in Figure 3.25 that α and β are generators since

$$\begin{array}{lll} \epsilon = \alpha^5 = \beta^2, & \beta_1 = \beta, & \beta_2 = \alpha^2\beta = \beta\alpha^3, \\ \beta_3 = \alpha^4\beta = \beta\alpha, & \beta_4 = \alpha\beta = \beta\alpha^4, & \beta_5 = \alpha^3\beta = \beta\alpha^2. \end{array}$$

In general, the automorphism group of the cycle C_n has order $2n$. This group is called a **dihedral group** and is commonly denoted by D_n. Thus the dihedral group D_5 has order 10 and its group table is shown in Figure 3.25. Furthermore, $D_3 = S_3$. $\diamondsuit$

A finite group A can have, indeed may require, a large generating set, although it is customary not to include the identity element of A as a generator. There is a digraph that one commonly associates with a group $A = \{a_1, a_2, \ldots, a_n\}$ and a generating set Δ for A. The **Cayley color digraph** $D_\Delta(A)$, named for the famous mathematician Arthur Cayley, has A as its vertex set, where (a_i, a_j), $i \neq j$, is an arc of $D_\Delta(A)$ if $a_j = a_i b$ for some generator



$\alpha = (v_1\ v_2\ v_3\ v_4\ v_5)$
$\beta_1 = (v_2\ v_5)(v_3\ v_4)$
$\beta_2 = (v_1\ v_3)(v_4\ v_5)$
$\beta_3 = (v_1\ v_5)(v_2\ v_4)$
$\beta_4 = (v_1\ v_2)(v_3\ v_5)$
$\beta_5 = (v_1\ v_4)(v_2\ v_3)$

	ϵ	α	α^2	α^3	α^4	β_1	β_2	β_3	β_4	β_5
ϵ	ϵ	α	α^2	α^3	α^4	β_1	β_2	β_3	β_4	β_5
α	α	α^2	α^3	α^4	ϵ	β_4	β_5	β_1	β_2	β_3
α^2	α^2	α^3	α^4	ϵ	α	β_2	β_3	β_4	β_5	β_1
α^3	α^3	α^4	ϵ	α	α^2	β_5	β_1	β_2	β_3	β_4
α^4	α^4	ϵ	α	α^2	α^3	β_3	β_4	β_5	β_1	β_2
β_1	β_1	β_3	β_5	β_2	β_4	ϵ	α^3	α	α^4	α^2
β_2	β_2	β_4	β_1	β_3	β_5	α^2	ϵ	α^3	α	α^4
β_3	β_3	β_5	β_2	β_4	β_1	α^4	α^2	ϵ	α^3	α
β_4	β_4	β_1	β_3	β_5	β_2	α	α^4	α^2	ϵ	α^3
β_5	β_5	β_2	β_4	β_1	β_3	α^3	α	α^4	α^2	ϵ

Figure 3.25: The group table of Aut(G) in Example 3.7

$b \in \Delta$. Furthermore, we label (or color) this arc by b. Consequently, every vertex a_i of $D_\Delta(A)$ is adjacent to the vertex $a_i b$ for each $b \in \Delta$. In addition, every vertex a_j of $D_\Delta(A)$ is adjacent from the vertex $a_j b^{-1}$ for each $b \in \Delta$. That is, the outdegree *and* the indegree of every vertex of $D_\Delta(A)$ is $|\Delta|$. We consider some examples of Cayley color digraphs.

Example 3.8 *For the Klein four group* $A = \{a_1, a_2, a_3, a_4\}$, *whose group table is given in Figure* 3.26, *the Cayley color digraph is shown in the same figure for the generating set* $\Delta = \{a_2, a_4\}$.

	a_1	a_2	a_3	a_4
a_1	a_1	a_2	a_3	a_4
a_2	a_2	a_1	a_4	a_3
a_3	a_3	a_4	a_1	a_2
a_4	a_4	a_3	a_2	a_1

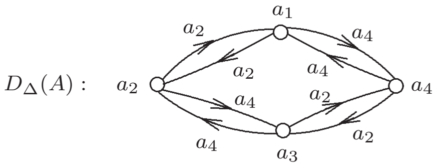


Figure 3.26: The group table and the Cayley color digraph in Example 3.8

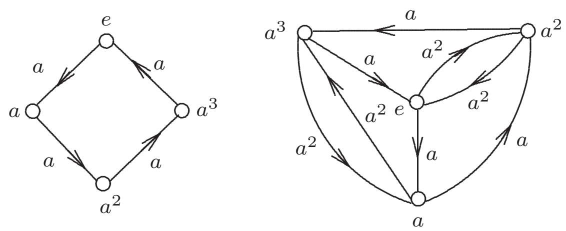
Example 3.9 *For the cyclic group* $A = \{e, a, a^2, a^3\}$, *whose group table is given in Figure* 3.27, *the Cayley color digraphs are shown in the same figure for the two generating sets* $\Delta_1 = \{a\}$ *and* $\Delta_2 = \{a, a^2\}$.

Just as every graph has an automorphism group, so too does every digraph. In the case of a Cayley color digraph D, those automorphisms ϕ of D that preserve colors (that is, such that the arcs (u, v) and $(\phi u, \phi v)$ have the same color (generator) for every arc (u, v) of D) form a subgroup of the automorphism group of the digraph D (whose arcs are not colored).

The topic of automorphism groups of graphs appeared in the first book on graph theory, in fact, early in the book. On page 5 of the first section (*The Basic Concepts*) of the first chapter (*Foundations*) of his 1936 book, Dénes König posed

	e	a	a^2	a^3
e	e	a	a^2	a^3
a	a	a^2	a^3	e
a^2	a^2	a^3	e	a
a^3	a^3	e	a	a^2

Group Table



$D_{\Delta_1}(A)$ $D_{\Delta_2}(A)$

Figure 3.27: The group table and Cayley color digraphs in Example 3.9

the following question (translated from German):

> *When can a given abstract group be interpreted as the group of a graph and if this is the case, how can the corresponding graph be constructed?*

In the early 1900s, Germany was known for its mathematicians who excelled in group theory. One of the best known was Ferdinand Georg Frobenius (1849–1917), who made numerous important contributions to several areas of mathematics, but especially to group theory. One of Frobenius's doctoral students was Issai Schur (1875–1941), who died on his 66th birthday (January 10).

Schur, a gifted mathematician, became well-known for his work in groups, particularly on representation theory, although Schur did research in many areas of mathematics. Indeed, through Schur's efforts, Berlin became known for the study of groups. Schur was very popular with students; often his lectures at the University of Berlin were given to overflow audiences. However, beginning in 1933, events in Germany made life very difficult for Schur, who was Jewish. In 1935 Schur was dismissed from his position at the University, and the remainder of his life was often unbearable.

While at the University of Berlin, Schur supervised several doctoral students, including Richard Rado, whom we will meet in Chapter 11, Richard Brauer (1901–1977), well-known for his work in algebra and number theory, and Helmut Wielandt (1910–2001), who made major contributions to the study of permutation groups and linear algebra. Another of Schur's students was Roberto Frucht (1906–1997).

Roberto Frucht entered the University of Berlin in 1924 at the age of 18. Although Frucht's favorite mathematical area was tensor calculus, he could not find a doctoral advisor in that area. Being an admirer of Schur, Frucht asked him if he would agree to be his advisor and Schur agreed – provided Frucht would write his thesis in an area of interest to Schur. Consequently, Frucht switched to group theory and received his Ph.D. in 1930.

At that time, Frucht's father lost his job and earning a living became a top priority for Frucht. Finding a job as a mathematician in Germany was difficult during that period, except as a high school teacher. However, German citizenship was required for that and Frucht was a Czechoslovakian citizen. Consequently, Frucht moved to Trieste, Italy to work at an Italian insurance company. Frucht stayed in Italy until 1938. During that period, his life was relatively inactive, mathematically speaking. However, one day in 1936 he received a catalog from Akademische Verlagsgesellschaft advertising a book in Graph Theory (by Dénes König). Frucht immediately ordered the book and he became an enthusiastic graph theorist the very day that the book arrived.

König's question on automorphism groups attracted the attention of Frucht immediately. After being unsuccessful for several months trying to solve the problem, he found a solution that seemed rather easy (after he had found it). In 1939 Frucht escaped from Italy to South America shortly before the outbreak of World War II. After working as an actuary in Argentina for awhile, he was successful in acquiring a position at the Universidad Santa Maria in Valparaiso, Chile. He continued his interest in graph theory and remained in Chile the rest of his life.

Theorem 3.10 (**Frucht's Theorem**) *For every finite group A, there exists a graph G such that* $\mathrm{Aut}(G)$ *is isomorphic to A.*

To prove Theorem 3.10, Frucht used his result that for every finite abstract group A, the group of color-preserving automorphisms of the Cayley color digraph $D_\Delta(A)$ is isomorphic to A. The next step for Frucht was to convert $D_\Delta(A)$ into a graph G so that the automorphisms of G correspond to the color-preserving automorphisms of $D_\Delta(A)$ in an appropriate manner.

Although this can be accomplished in a number of ways, one way is to replace each arc (u, v) labeled $b_1 \in \Delta$ in $D_\Delta(A)$ by a path u, u_1, v_1, v, where u_1 and v_1 are new vertices, and place a pendant edge at u_1 and attach a path of length 2 at v_1. If there is another generator, say b_2, then replace any arc (u, v) labeled b_2 in $D_\Delta(A)$ by a path u, u_2, v_2, v and attach a path of length 3 at u_2 and a path of length 4 at v_2. This procedure is continued if there are additional generators. See Figure 3.28.

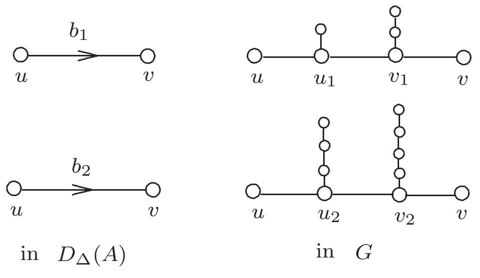


Figure 3.28: Constructing a graph G from $D_\Delta(A)$

For example, consider the Cayley color digraph $D_\Delta(A)$ in Example 3.8, which is redrawn in Figure 3.29. The associated graph G is also shown in that figure, where $b_1 = a_2$ and $b_2 = a_4$. By Frucht's theorem, $\mathrm{Aut}(G)$ is isomorphic to A (the Klein four group).

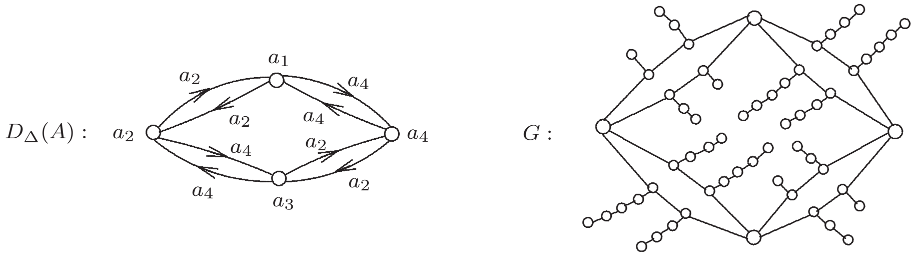


Figure 3.29: Constructing a graph G from $D_\Delta(A)$

Exercises for Section 3.3

3.20 For the graph H of Figure 3.19, give an example of a permutation on $V(H)$ that preserves degree but which is not an automorphism of H.

3.21 Determine the automorphism group of K_3.

3.22 Determine the automorphism group of $K_{1,3}$.

3.23 Determine the automorphism group of P_n for $n \geq 2$.

3.24 Determine the automorphism group of C_4.

3.25 For each of the graphs H_1 and H_2 in Figure 3.30, determine

(a) the orbits of the graph.

(b) the automorphism group of the graph.

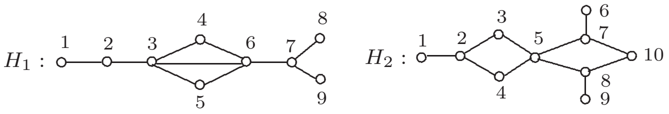


Figure 3.30: The graphs in Exercise 3.25

3.26 For the graph G_2 in Figure 3.23, determine

(a) the orbits of G_2.

(b) the automorphism group of G_2.

3.27 Prove that $\mathrm{Aut}(G)$ and $\mathrm{Aut}(\overline{G})$ are isomorphic for every graph G.

3.28 For the group $A = \{e, a, b\}$ whose group table is given in Figure 3.31,

(a) find a generating set Δ and the corresponding Cayley color digraph.

(b) use the Cayley color digraph in (a) to construct a graph G such that $\mathrm{Aut}(G) \cong A$.

	e	a	b
e	e	a	b
a	a	b	e
b	b	e	a

Figure 3.31: The group table in Exercise 3.28

(c) find a graph H of order 12 such that $\mathrm{Aut}(H) \cong A$.

3.29 Consider the group $A = \{e, a, b, c\}$ whose group table is given in Figure 3.32, where $a^2 = b^2 = c^2 = e$.

(a) For $\Delta = \{a, b\}$, find the corresponding Cayley color digraph.

(b) Use the Cayley color digraph in (a) to construct a graph G such that $\mathrm{Aut}(G) \cong A$.

	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

Figure 3.32: The group table in Exercise 3.29

3.30 Consider the group $A = \{e, a, b, c, d\}$ whose group table is given in Figure 3.33.

(a) For $\Delta = \{a\}$, find the corresponding Cayley color digraph.

(b) Use the Cayley color digraph in (a) to construct a graph G such that $\mathrm{Aut}(G) \cong A$.

	e	a	b	c	d
e	e	a	b	c	d
a	a	b	c	d	e
b	b	c	d	e	a
c	c	d	e	a	b
d	d	e	a	b	c

Figure 3.33: The group table in Exercise 3.30

3.31 Let $A = S_3$ be the symmetric group on the set $\{1, 2, 3\}$. For $\Delta = \{(123), (12)\}$,

(a) find the corresponding Cayley color digraph,

(b) use the Cayley color digraph in (a) to construct a graph G such that $\mathrm{Aut}(G) \cong A$.

3.32 Use each of the Cayley color digraphs in Example 3.9 to construct a graph G such that $\mathrm{Aut}(G) \cong A$.

3.4 Excursion: Reconstruction and Solvability

Figure 3.34 shows a deck of five cards, each with a drawing of a graph.

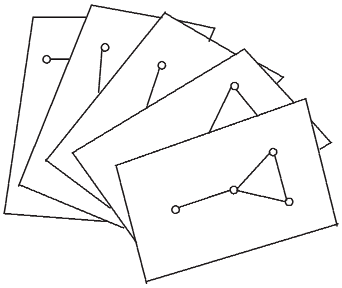


Figure 3.34: A deck of cards

Since we can't see the graph drawn on each card, we separate the cards, which are shown in Figure 3.35. We've also numbered the cards now. What do you notice about the graphs on these five cards? Of course, the graphs on cards 1 and 2 are isomorphic, as are the graphs on cards 3 and 4. However, the observation we are looking for is that all graphs have order 4.

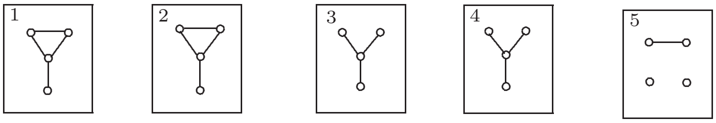


Figure 3.35: A deck of five cards

It turns out that there exists a certain graph G of some order n such that for each $v \in V(G)$, the unlabeled subgraph $G - v$ is drawn on one of the cards in the deck given in Figure 3.35. We can see that $n = 5$ in two ways. First, n must equal 5 since there are five cards. Also, each subgraph $G - v$, $v \in V(G)$, has order $n - 1 = 4$ and so $n = 5$.

If the value of a parameter for a graph G or whether a graph G has a certain property can be determined from the (unlabeled) graphs $G - v$, $v \in V(G)$, then this parameter or property is said to be **recognizable** for G. The observations we have made above provide a proof in general for the following.

Theorem 3.11 *The order of every graph is recognizable.*

Before continuing, let's look at the (very small) deck of cards in Figure 3.36. Since there are only two cards in the deck and each card is the (trivial) graph of order 1, the graph G in question has order 2. Actually, there are two (non-isomorphic) graphs of order 2, namely K_2 and $\overline{K}_2$. Of course, the size of K_2 is 1 and the size of $\overline{K}_2$ is 0. But, whether $G \cong K_2$ or $G \cong \overline{K}_2$, the two graphs $G - v$ in each case are K_1. That is, if $G \cong K_2$ or $G \cong \overline{K}_2$, then there is no way to determine the size of G from the subgraphs $G - v$, $v \in V(G)$. Therefore, the sizes of K_2 and $\overline{K}_2$ are not recognizable. This is the exception, however, rather than the rule.

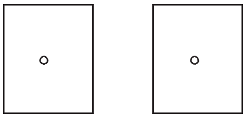


Figure 3.36: A deck of two cards

Theorem 3.12 *The size of every graph of order at least* 3 *is recognizable.*

Proof. [direct proof] Suppose that G is a graph of order $n \geq 3$ and size m. Let $V(G) = \{v_1, v_2, \ldots, v_n\}$. Of course, in the deck consisting of the subgraphs $G - v_i$, $1 \leq i \leq n$, the vertices will not be labeled. Let e be an edge of G, say $e = v_1v_2$. The edge e then appears in each of the subgraphs $G - v_i$ for $3 \leq i \leq n$ but appears in neither $G - v_1$ nor $G - v_2$. Suppose that the size of $G - v_i$ is m_i $(1 \leq i \leq n)$. In the sum $\sum_{i=1}^{n} m_i$, every edge is counted $n - 2$ times. That is,

$$\sum_{i=1}^{n} m_i = m(n-2) \quad \text{and so} \quad m = \frac{\sum_{i=1}^{n} m_i}{n-2}. \qquad \blacksquare$$

Let card i $(1 \leq i \leq 5)$ in Figure 3.35 display the subgraph $G - v_i$, which has size m_i. Thus

$$\sum_{i=1}^{5} m_i = 4 + 4 + 3 + 3 + 1 = 15.$$

By Theorem 3.12, $m = 15/(5-2) = 5$. Hence the size of the graph G in question described by the deck in Figure 3.35 is 5.

Now that we know that the size of every graph G of order at least 3 is recognizable, we can show that one additional feature of G is recognizable.

Theorem 3.13 *A degree sequence of every graph of order at least* 3 *is recognizable.*

Proof. [direct proof] Let G be a graph of order $n \geq 3$ and size m, where $V(G) = \{v_1, v_2, \ldots, v_n\}$. From Theorem 3.12, m can be determined from the subgraphs $G - v_i$ $(1 \leq i \leq n)$. Suppose that the size of $G - v_i$ is m_i for $1 \leq i \leq n$. That is, the size of G is m but when the vertex v_i is removed from G, the size of the resulting subgraph $G - v_i$ is m_i. Consequently, $\deg v_i = m - m_i$ and so $m - m_1, m - m_2, \ldots, m - m_n$ is a degree sequence for G. ■

Returning to the subgraphs $G - v_i$ $(1 \leq i \leq 5)$ shown on the deck of the cards in Figure 3.35, we see that $m_1 = m_2 = 4$, $m_3 = m_4 = 3$, and $m_5 = 1$. Since we have already seen that the size m of G is 5, it follows that $1, 1, 2, 2, 4$, is a degree sequence of G. In particular, $\deg v_5 = 4$ so that v_5 is adjacent to each of the four vertices of the graph $G - v_5$ given in card 5. Hence we now know precisely (up to isomorphism) what the mystery graph G is for the deck of cards in Figure 3.35. The graph G is shown in Figure 3.37.

G :

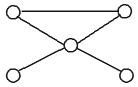


Figure 3.37: The graph G for the deck of cards in Figure 3.35

Consequently, from the deck of cards in Figure 3.35 that gives the subgraphs $G - v_i$ $(1 \leq i \leq 5)$, we have not only been able to determine the order of G, the size of G, and a degree sequence for G, we have been able to determine G itself.

A graph G of order $n \geq 2$ is **reconstructible** if G can be uniquely determined (up to isomorphism) from its subgraphs $G - v_i$ $(1 \leq i \leq n)$. Thus, the graph G of Figure 3.37 is reconstructible. From our earlier remarks, neither graph of order 2 is reconstructible. It is believed by many but has not been verified by any that every graph of order 3 or more is reconstructible.

The Reconstruction Conjecture *Every graph of order* 3 *or more is reconstructible.*

This conjecture is believed to have been made in 1941 and is often attributed jointly to Paul J. Kelly (1915-1995) and Stanislaw M. Ulam. Kelly, who spent many years as a faculty member at the University of California at Santa Barbara, obtained a number of results on this topic. Ulam was born on April 3, 1909 in Lemberg, Poland (now Lvov, Ukraine). He became interested in astronomy, physics, and mathematics while a teenager and learned calculus on his own. He entered the Polytechnic Institute in Lvov in 1927. One of his professors there was Kazimierz Kuratowski, whom we will visit again in Chapter 9. Ulam studied under Stefan Banach and received his Ph.D. in 1933.

In 1940, Ulam acquired a position as an assistant professor at the University of Wisconsin (where Kelly was studying for his Ph.D.). Three years later, John von Neumann asked to meet Ulam at a railroad station in Chicago. This led to Ulam going to the Los Alamos National Laboratory in New Mexico to work on the hydrogen bomb with the physicist Edward Teller. While at Los Alamos, Ulam developed the well-known Monte Carlo method for solving mathematical problems using a statistical sampling method with random numbers.

Throughout his life, he made important contributions in many areas of mathematics. Ulam died on May 13, 1984.

The Reconstruction Problem is to determine whether the Reconstruction Conjecture is true. Solving the Reconstruction Problem requires verifying that there do not exist two nonisomorphic graphs G and H such that the set of subgraphs $G-v$, $v \in V(G)$, and the set of subgraphs $H-v$, $v \in V(H)$, are the same. If there are two nonisomorphic graphs G and H with this property, then these graphs must have the same order, the same size, and the same degree sequence, as all of these parameters and properties are recognizable. Furthermore, any two such graphs G and H must both be connected or must both be disconnected. This is a consequence of Theorem 1.10, which states that a graph G of order 3 or more is connected if and only if G contains two distinct vertices u and v such that $G-u$ and $G-v$ are connected.

Theorem 3.14 *For all graphs of order at least* 3*, both connectedness and disconnectedness are recognizable properties.*

Proof. [direct proof] By Theorem 1.10, at least two of the subgraphs $G-v$, $v \in V(G)$, are connected if and only if G is connected. ■

The Reconstruction Problem concerns providing some information about a certain graph (or certain graphs) and asks us to show that only one graph G satisfies the given information, as well as to identify what this graph G is. In this case, the given information is a collection of subgraphs of G, namely all subgraphs of the type $G-v$, where $v \in V(G)$. However, we could be given a wide variety of information concerning a graph. Proceeding in a manner we discussed earlier, let's assume that we are given pieces of information concerning a graph G, where each item is written on a card. The set of all such cards is our **deck**. Any graph G that satisfies all information in the deck is called a **solution of the deck**. The question then becomes to determine all solutions of the deck. A deck may then have a unique solution, two or more solutions, or no solution. If, for a given graph G of order $n \geq 3$, a deck consists of all n subgraphs of the type $G-v$, where $v \in V(G)$, and if the the Reconstruction Conjecture is true, then the deck has a unique solution, namely G.

Example 3.15 *Find all graphs G for which the deck of cards shown in Figure* 3.38 *gives the subgraphs $G-v$, where $v \in V(G)$.*

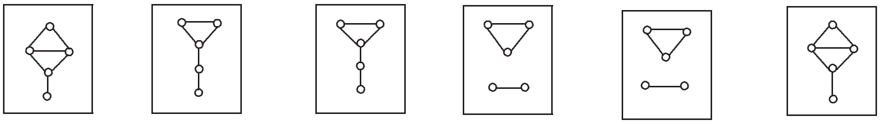


Figure 3.38: The deck of cards for Example 3.15

Solution. First, observe that the order of any solution G of this deck is 6. The sum of the sizes of the graphs on the deck is $6+5+5+4+4+6=30$. Then the size of a solution G is $30/(6-2) = 7.5$. This is impossible and so the graphs on the deck in Figure 3.38 are not the subgraphs $G - v$, $v \in V(G)$, of any graph G. Thus this deck has no solution. $\diamondsuit$

Example 3.16 *Determine the solutions of the deck of cards shown in Figure* 3.39.

Card #1	Card #2	Card #3	Card #4	Card #5
All subgraphs $G - v$ contain at most two triangles	Exactly one subgraph $G - v$ contains exactly two triangles	Some subgraph $G - v$ contains no triangles	Some subgraph $G - v$ is disconnected	$\Delta(G) \geq 5$

Figure 3.39: The deck of cards for Example 3.16

Solution. Suppose that G is a solution. By Card #1, all subgraphs $G - v$ of a solution G contain at most two triangles, and by Card #2 exactly one subgraph $G - v$, say $G - v_1$, contains exactly two triangles. The two triangles in $G - v_1$ are either disjoint, have one vertex in common, or have an edge in common, as in Figure 3.40.

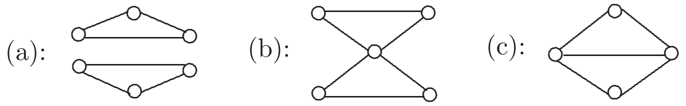


Figure 3.40: Possible subgraphs of a solution

We can eliminate the subgraph in Figure 3.40(a) because of Card #3. If the subgraph in Figure 3.40(c) occurs, then by Card #5, G must contain at least two additional vertices x and y. In that case, both $G - x$ and $G - y$ contain at least two triangles, contradicting Card #2. Necessarily then G contains the subgraph in Figure 3.40(b) with one additional vertex z. Furthermore, this subgraph must be an induced subgraph since $G - v_1$ has exactly two triangles. Certainly, z must be adjacent to the vertex of degree 4 because of Card #5. If z is adjacent to any other vertices in the subgraph in Figure 3.40(b), then Card #2 is contradicted. Hence we arrive at only one graph G, namely, the graph G of Figure 3.41. Card #4 is not needed. (It was a Joker!) $\diamondsuit$

Exercises for Section 3.4

3.33 Give an example of two nonisomorphic graphs G and H of order 3 or more containing vertices u and v, respectively, such that $G - u$ and $H - v$ are isomorphic, or explain why no such example exists.

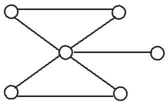


Figure 3.41: The unique solution to the deck in Figure 3.39

3.34 For the deck D of cards given in Figure 3.42, where card i contains the subgraph $G_i \cong G - v_i$, $v_i \in V(G)$, for some graph G, answer the following with explanation.

(a) What is the order n of G?

(b) What is the size m of G?

(c) What are the degrees of the vertices of G?

(d) Is G connected?

(e) What are the solutions of D?

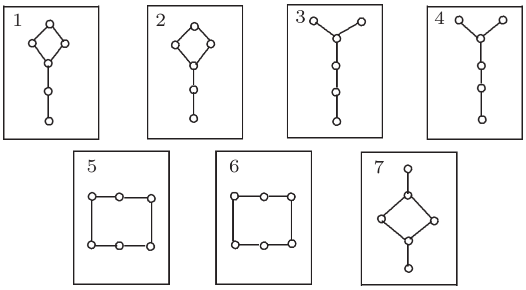


Figure 3.42: The deck of cards for Exercise 3.34

3.35 For a graph G of order n and size m, the subgraphs $G - v$ for $v \in V(G)$ are given on the deck of cards in Figure 3.43. Answer the following with explanation.

(a) What is n?

(b) What is m?

(c) Is G connected?

(d) What is a degree sequence of G?

(e) Find all solutions of the deck.

3.36 Determine the solutions G of the deck of cards shown in Figure 3.44.

3.37 Determine the solutions of the deck of cards shown in Figure 3.45.

3.38 Determine the solutions of the deck of two cards shown in Figure 3.46.

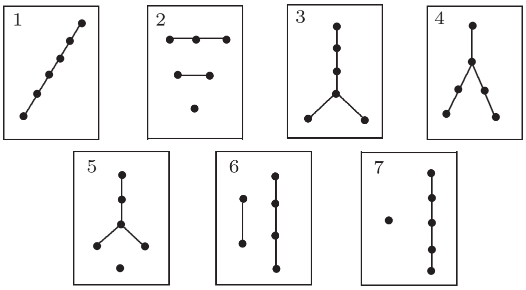


Figure 3.43: The deck of cards for Exercise 3.35

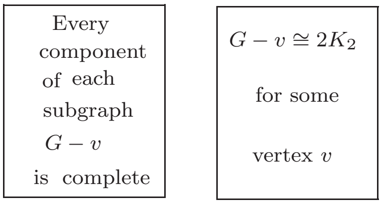


Figure 3.44: The deck of cards for Exercise 3.36

3.39 Determine the solutions of the deck of two cards shown in Figure 3.47.

3.40 Determine the solutions of the deck of one card shown in Figure 3.48.

3.41 Determine the solutions of the deck D of two cards shown in Figure 3.49.

3.42 Let $G = 3K_2 + K_1$.

(a) Give an example of a deck of cards for which G is the unique solution. Show your work.

(b) Find a deck with a small number of cards for which G is the unique solution. Show your work.

(c) Give an example of a deck D of cards for which G and one other graph are the only two solutions of D. Show your work.

3.43 Give an example of a deck of cards having exactly two solutions having order 3 or more.

3.44 Give an example of a deck of three cards having no solution, where any subdeck consisting of two of the three cards has at least one solution.

G is connected	$\Delta(G-v) \leq 2$ for every vertex v of G	There is no subgraph $G-v$ with $\Delta(G-v) = 1$	$\Delta(G-v) = 2$ for some vertex v of G

Figure 3.45: The deck of cards for Exercise 3.37

For every two distinct vertices u and v of G, the sizes of $G-u$ and $G-v$ are different	Some subgraph $G-v$ has order 5

Figure 3.46: The deck of cards for Exercise 3.38

only one subgraph $G-v$ contains exactly two vertices that are not adjacent	Two different subgraphs $G-v$ contain only odd vertices

Figure 3.47: The deck of cards for Exercise 3.39

Every subgraph $G-v$, where $v \in V(G)$, is regular

Figure 3.48: The deck of cards for Exercise 3.40

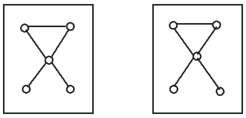


Figure 3.49: The deck of cards for Exercise 3.41

Chapter 4

Trees

4.1 Bridges

Suppose that there are some villages in a sparsely populated region where country roads allow us to travel directly between certain pairs of these villages. Since the traffic along these roads is ordinarily light, it is not surprising that very few roads have been built in this region. In fact, suppose that we have the situation illustrated in Figure 4.1, where there are seven villages denoted by $v_1, v_2, \ldots, v_7$ and six roads. Not only can this map be modeled by the graph G of Figure 4.1, the map essentially *is* a graph.

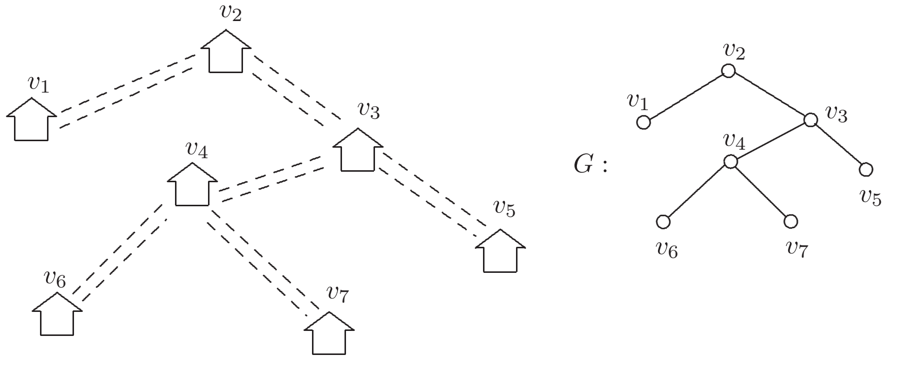


Figure 4.1: A graph model of villages and roads

There are two interesting features of the map (and the graph) of Figure 4.1. First, you may have heard of the traveler who stops someplace during his trip and asks a local resident for directions to some location: "How do you get there?" only to get the response "You can't get there from here." Well, fortunately, we don't have *that* situation with the villages in Figure 4.1. Indeed, it is possible to travel along country roads from each of the seven villages to all other villages. In other words, the graph G of Figure 4.1 is connected. Although this is a very positive feature (an essential feature, one might say), the map and graph also have a negative feature. Namely, if it ever became necessary to close any of the roads due to road construction, flooding, or a major snowstorm, then it would no longer be possible to travel between every two villages. In terms of the graph G of Figure 4.1, this says that if we were to remove any edge of G, then the resulting graph would no longer be connected. An edge with this property plays an important role in graph theory.

Recall that if e is an edge of a graph G, then $G - e$ is the subgraph of G having the same vertex set as G and whose edge set consists of all edges of G except e. Also, if X is a set of edges of G, then $G - X$ is the subgraph possessing the same vertex set as G and all edges of G except those in X. If G has order n, then $G - E(G)$ is the empty graph $\overline{K}_n$.

An edge $e = uv$ of a connected graph G is called a **bridge** of G if $G - e$ is disconnected. In this case, $G - e$ necessarily contains exactly two components, one containing u and the other containing v. If the vertex v, say, has degree 1, then the component of $G - e$ containing v is a single vertex, in which case $G - v$ has only one component; that is, if v is an end-vertex of a connected graph G, then $G - v$ is connected. An edge e is a **bridge** of a disconnected graph if e is a bridge of some component of G. In the disconnected graph G of Figure 4.2, the edges u_2u_5, v_3v_4, v_4v_5, and w_1w_2 are bridges (which are indicated in bold). No other edges of G are bridges.

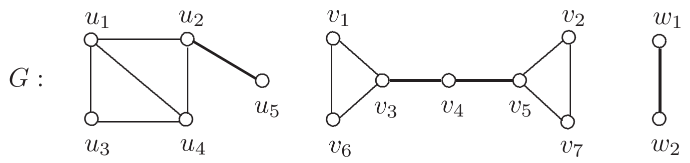


Figure 4.2: A disconnected graph with four bridges

The following theorem makes it easy to determine which edges in a graph are bridges.

Theorem 4.1 *An edge e of a graph G is a bridge if and only if e lies on no cycle of G.*

Proof. [proof by contrapositive] Let $e = uv$ be an edge of G that is not a bridge. Suppose that e lies in the component G_1 of G. (Of course $G_1 = G$ if G is connected.) Then $G_1 - e$ is connected. Hence there exists a $u - v$ path P in $G_1 - e$; however, P together with e form a cycle containing e in G_1 and therefore in G as well.

We now verify the converse. Suppose that $e = uv$ lies on a cycle C of G and that e (and C) belong to the component G_1 of G. Then there is a $u - v$ path P' in G_1 not containing e. We show that $G_1 - e$ is connected. Let x and y be any two vertices of $G_1 - e$. We show that x and y are connected in $G_1 - e$. Since G_1 is connected, G_1 contains an $x - y$ path Q. If e is not on Q, then Q is an $x - y$ path in $G_1 - e$ as well. On the other hand, if e lies on Q, then replacing e in Q by the $u - v$ path P' produces an $x - y$ walk. By Theorem 1.6, $G_1 - e$ contains an $x - y$ path. ■

The graph G of Figure 4.1 is connected and has no cycles. Therefore, every edge of G is a bridge. Graphs with these two properties are especially important and will be the main subject of this chapter.

Exercises for Section 4.1

4.1 Give an example of a nontrivial connected graph G with the properties that (1) every bridge of G is adjacent to an edge that is not a bridge, (2) every edge of G that is not a bridge is adjacent to a bridge, (3) G contains two nonadjacent bridges, and (4) every two edges of G that are not bridges are adjacent.

4.2 Prove that every connected graph all of whose vertices have even degrees contains no bridges.

4.3 Prove that if uv is a bridge in a graph G, then there is a unique $u - v$ path in G.

4.4 Let G be a connected graph, and let e_1 and e_2 be two edges of G. Prove that $G - e_1 - e_2$ has three components if and only if both e_1 and e_2 are bridges in G.

4.5 (a) Let G be a connected graph of order n, where every edge of G is a bridge. What is the size of G? Explain.

(b) Let G be a disconnected graph of order n having k components, where every edge of G is a bridge. What is the size of G? Explain.

4.6 Let G be a connected graph of order $n \geq 3$ without bridges. Suppose that for every edge e of G, each edge of $G - e$ is a bridge. What is G? Justify your answer.

4.2 Trees

A graph G is called **acyclic** if it has no cycles. A **tree** is an acyclic connected graph. Therefore, the graph G of Figure 4.1 is a tree. When dealing with trees, we often use T rather than G to denote a tree. By Theorem 4.1, every edge in a tree is a bridge. Indeed, we could define a tree as a connected graph, every edge of which is a bridge. Figure 4.3 shows all six trees of order 6. The tree $T_1 \cong K_{1,5}$ is a star and $T_6 \cong P_6$ is a path. The number of end-vertices in the trees of Figure 4.3 ranges from 2 to 5. We'll have more to say about this shortly. A tree containing exactly two vertices that are not end-vertices (which are necessarily adjacent) is called a **double star**. The trees T_2 and T_3 in Figure 4.3 are double stars.

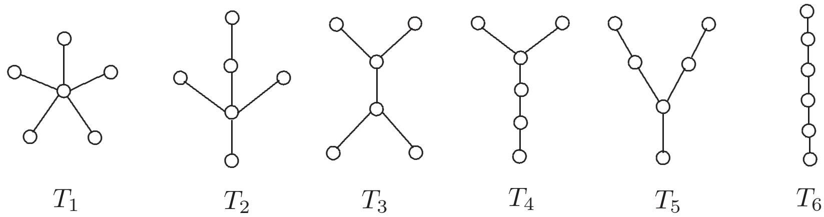


Figure 4.3: The trees of order 6

Another common class of trees consists of the "caterpillars". A **caterpillar** is a tree of order 3 or more, the removal of whose end-vertices produces a path called the **spine** of the caterpillar. Thus every path and star (of order at least 3) and every double star is a caterpillar, as is every tree shown in Figure 4.3. The trees T' and T'' of Figure 4.4 are also caterpillars, but T''' is not.

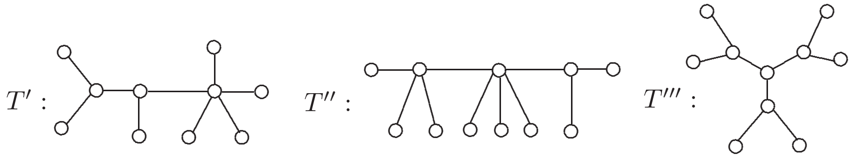


Figure 4.4: Two caterpillars and a tree that is not a caterpillar

There are occasions when it is convenient to select a vertex of a tree T under discussion and designate this vertex as the **root** of T. The tree T then becomes a **rooted tree**. Often the rooted tree T is drawn with the root r at the top and the other vertices of T drawn below, in levels, according to their distances from r. An example is given in Figure 4.5.

Given a labeled tree T with distinct vertices r and s, the results of rooting T at r and rooting T at s are considered different rooted trees (even if there is an isomorphism from T to itself that maps r to s).

Acyclic graphs are also referred to as **forests**. Therefore, each component of a forest is (not surprisingly) a tree. Of course, the one fact that distinguishes trees from forests is that a tree is required to be connected, while a forest is not required to be connected. Since a tree is connected, every two vertices in a tree are connected by a path. In fact, we can say more.

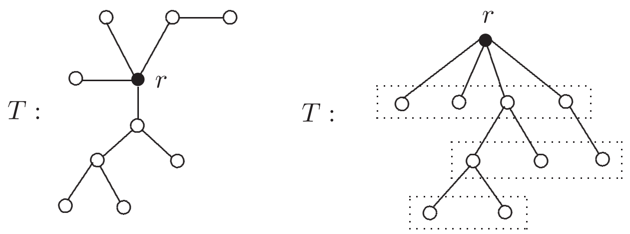


Figure 4.5: A rooted tree

Theorem 4.2 *A graph G is a tree if and only if every two vertices of G are connected by a unique path.*

Proof. [proof by contradiction] First, let G be a tree. Then G is connected by definition. Thus every two vertices of G are connected by a path. Assume, to the contrary, that there are two vertices of G that are connected by two distinct paths. Then a cycle is produced from some or all of the edges of these two paths. This is a contradiction.

For the converse, suppose that every two distinct vertices of G are connected by a unique path. Certainly then, G is connected. Assume, to the contrary, that G has a cycle C. Let u and v be two distinct vertices of C. Then C determines two distinct $u - v$ paths, producing a contradiction. Thus G is acyclic and so G is a tree. ■

As we have already observed, each tree in Figures 4.3 and 4.4 has two or more end-vertices. All nontrivial trees have this property.

Theorem 4.3 *Every nontrivial tree has at least two end-vertices.*

Proof. [direct proof] Let T be a nontrivial tree and among all paths in T, let P be a path of greatest length. Suppose that P is a $u - v$ path, say $P : u = u_0, u_1, \ldots, u_k = v$, where $k \geq 1$. We show that u and v are end-vertices of G. Necessarily, neither u nor v is adjacent to any vertex not on P, for otherwise, a path of greater length would be produced. Certainly, u is adjacent to u_1 on P and v is adjacent to u_{k-1} on P. Moreover, since T contains no cycles, neither u nor v is adjacent to any other vertices in P. Therefore, $\deg u = \deg v = 1$. ■

One major consequence of this result is that if T is a tree of order $k+1 \geq 2$, then for each end-vertex v of T, the subgraph $T - v$ is a tree of order k. This fact is useful for induction proofs of results concerning trees. We illustrate this idea now by showing that the size of every tree is one less than its order, another useful property of trees.

Theorem 4.4 *Every tree of order n has size $n - 1$.*

Proof. [induction] We proceed by induction on n. There is only one tree of order 1, namely K_1, which has size 0. Thus the result is true for $n = 1$. Assume for a positive integer k that the size of every tree of order k is $k-1$. Let T be a tree of order $k+1$. By Theorem 4.3, T contains at least two end-vertices. Let v be one of them. Then $T' = T - v$ is a tree of order k. By the induction hypothesis, the size of T' is $m = k-1$. Since T has exactly one more edge than T', the size of T is $m + 1 = (k-1) + 1 = (k+1) - 1$, as desired. ■

Let's illustrate some of the ideas that we've just discussed.

Example 4.5 *The degrees of the vertices of a certain tree T of order 13 are $1, 2,$ and 5. If T has exactly three vertices of degree 2, how many end-vertices does it have?*

Solution. Since T has three vertices of degree 2, it has ten vertices of degree 1 or 5. Let x denote the number of end-vertices of T. So T has $10 - x$ vertices of degree 5. Since T has 13 vertices, T has 12 edges by Theorem 4.4. Summing the degrees of the vertices of T and applying the First Theorem of Graph Theory, we obtain

$$\begin{aligned} 1 \cdot x + 2 \cdot 3 + 5 \cdot (10 - x) &= 2 \cdot 12 \\ x + 6 + 50 - 5x &= 24 \\ x &= 8. \quad \diamondsuit \end{aligned}$$

Note that drawing a tree of order 13 with three vertices of degree 2, two vertices of degree 5, and eight end-vertices does not answer the question. It only says that the tree we drew has eight end-vertices, not necessarily that the tree T in Example 4.5 has eight end-vertices. Of course, *our* solution tells us that *every* tree with the property described in Example 4.5 has eight end-vertices.

We can now determine the size of a forest in terms of its order and the number of components it has.

Corollary 4.6 *Every forest of order n with k components has size $n - k$.*

Proof. [direct proof] Suppose that the size of a forest F is m. Let $G_1, G_2, \ldots, G_k$ be the components of F, where $k \geq 1$. Furthermore, suppose that G_i has order n_i and size m_i for $1 \leq i \leq k$. Then $n = \sum_{i=1}^{k} n_i$ and $m = \sum_{i=1}^{k} m_i$. Since each component G_i $(1 \leq i \leq k)$ is a tree, it follows by Theorem 4.4 that $m_i = n_i - 1$. Therefore,

$$m = \sum_{i=1}^{k} m_i = \sum_{i=1}^{k} (n_i - 1) = n - k. \qquad ■$$

Again by Theorem 4.4, a tree of order n is a connected graph containing $n-1$ edges. Indeed, every connected graph of order n contains at least $n - 1$ edges. Although this can be verified in several ways, we establish this fact using a proof by minimum counterexample in order to illustrate this useful method of proof. (Proof by minimum counterexample is reviewed in Appendix 3.)

Theorem 4.7 *The size of every connected graph of order n is at least $n-1$.*

Proof. [proof by minimum counterexample] It is not difficult to see that the theorem is true for connected graphs of order 1, 2, or 3. Assume that the theorem is false however. Then there exists a connected graph G of smallest order n whose size m is at most $n-2$. Necessarily, $n \geq 4$. Since G is a nontrivial connected graph, G contains no isolated vertices.

We claim that G contains an end-vertex; for assume, to the contrary, that the degree of every vertex of G is at least 2. Then the sum of the degrees of the vertices of G is $2m \geq 2n$; so $m \geq n \geq m+2$, which is impossible. So, as we claimed, G contains an end-vertex.

Let v be an end-vertex of G. Since G is connected, has order n, and size $m \leq n-2$, it follows that $G-v$ is connected and has order $n-1$ and size $m-1 \leq n-3$, contradicting the assumption that G is a connected graph of smallest order whose size is at least 2 less than its order. ∎

Let G be a tree of order n and size m. By the definition of a tree and Theorem 4.4, G has the following three properties: (1) G is connected, (2) G is acyclic, (3) $m = n-1$. In fact, if G is a graph of order n and size m that satisfies *any* two of these three properties, then G is a tree.

Theorem 4.8 *Let G be a graph of order n and size m. If G satisfies any two of the properties:*

(1) *G is connected,* (2) *G is acyclic,* (3) $m = n-1$,

then G is a tree.

Proof. [proof by contradiction, direct proof] First, if G satisfies (1) and (2), then G is a tree by definition. Thus, we may assume that G satisfies (1) and (3) or G satisfies (2) and (3). We consider these two cases.

Case 1. *G satisfies* (1) *and* (3). Since G is connected, it suffices to show that G is acyclic. Assume, to the contrary, that G contains a cycle C. Let e be an edge of C. Then e is not a bridge of G by Theorem 4.1. So $G-e$ is a connected graph of order n and size $n-2$, which contradicts Theorem 4.7. Therefore, G is acyclic and so is a tree.

Case 2. *G satisfies* (2) *and* (3). Since G is acyclic, it suffices to show that G is connected. Since G satisfies (2) and (3), it follows that G is a forest of order n and size $m = n-1$. By Corollary 4.6, the size of G is $n-k$, where k is the number of components of G. Hence $n-1 = n-k$ and so $k = 1$. Therefore, G is connected. ∎

If T is a tree of order k, then it should be clear that T is isomorphic to a subgraph of K_k. Of course, $\delta(K_k) = k-1$. Not only is T isomorphic to a subgraph of K_k, the tree T is isomorphic to a subgraph of *every* graph having minimum degree at least $k-1$.

Theorem 4.9 *Let T be a tree of order k. If G is a graph with $\delta(G) \geq k-1$, then T is isomorphic to some subgraph of G.*

Proof. [induction] We proceed by induction on k. The result is certainly true for $k = 1$ since every graph contains a vertex. It is also true for $k = 2$ since every graph without isolated vertices contains edges.

Assume for every tree T' of order $k-1$, where $k \geq 3$, and for every graph H with minimum degree at least $k-2$ that T' is isomorphic to a subgraph of H. Now, let T be a tree of order k and let G be a graph with $\delta(G) \geq k-1$. We show that T is isomorphic to a subgraph of G.

Let v be an end-vertex of T and u the vertex of T adjacent to v. Then $T-v$ is a tree of order $k-1$. Since $\delta(G) \geq k-1 \geq k-2$, it follows by the induction hypothesis that $T-v$ is isomorphic to a subgraph F of G. Let u' denote the vertex of F corresponding to u in T. Since $\deg_G u' \geq k-1$ and the order of F is $k-1$, the vertex u' is adjacent to a vertex w of G that does not belong to F (see Figure 4.6). Therefore, T is isomorphic to a subgraph of G. ■

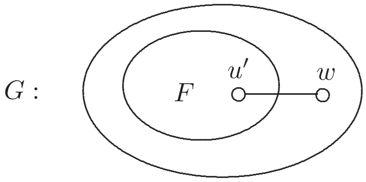


Figure 4.6: The subgraph $T-v$ of G in the proof of Theorem 4.9

Exercises for Section 4.2

4.7 (a) Draw all trees of order 5.

(b) Draw all forests of order 6.

4.8 Prove that if every vertex of a graph G has degree at least 2, then G contains a cycle.

4.9 Show that a graph of order n and size $n-1$ need not be a tree.

4.10 Give an example of each of the following or explain why no such example exists.

(a) A graph that is not a tree in which every edge is a bridge.

(b) A tree of order 4 whose complement is not a tree.

(c) A tree T containing exactly three vertices that are not end-vertices and such that T is not a caterpiller.

4.11 For $k = 2, 3, 4$, give an example of a tree T_k with $\Delta(T_k) = k$ such that no two vertices of the same degree are adjacent to the same vertex.

4.12 (a) Give an example of a tree T and an edge e of T such that the two components of $T - e$ are isomorphic.

(b) Show that there exists no tree T containing two distinct edges e_1 and e_2 such that the two components of $T - e_1$ are isomorphic *and* the two components of $T - e_2$ are isomorphic.

4.13 A certain tree T of order 21 has only vertices of degree 1, 3, 5, and 6. If T has exactly 15 end-vertices and one vertex of degree 6, how many vertices of T have degree 5?

4.14 A certain tree T of order 35 is known to have 25 vertices of degree 1, two vertices of degree 2, three vertices of degree 4, one vertex of degree 5, and two vertices of degree 6. It also contains two vertices of the same (unknown) degree x. What is x?

4.15 A tree T with 50 end-vertices has an equal number of vertices of degree 2, 3, 4, and 5 and no vertices of degree greater than 5. What is the order of T?

4.16 (a) Give an example of a tree of order 6 containing four vertices of degree 1 and two vertices of degree 3. (Only one tree has these properties.)

(b) Find all trees T where two-thirds of the vertices of T have degree 1 and the remaining one-third of the vertices have degree 3.

4.17 (a) Give an example of a tree of order 8 containing six vertices of degree 1 and two vertices of degree 4. (Only one tree has these properties.)

(b) Find all trees T where 75% of the vertices of T have degree 1 and the remaining 25% of the vertices have degree 4.

(c) Find all trees T where 75% of the vertices of T have degree 1 and the remaining 25% of the vertices have another degree (a fixed degree).

(d) Find all trees T where 25% of the vertices of T have degree 1 and the remaining 75% of the vertices of T have another degree (a fixed degree).

4.18 A certain tree T of order n contains only vertices of degree 1 and 3. Show that T contains $(n-2)/2$ vertices of degree 3.

4.19 Let T be a tree of order n and size m having n_i vertices of degree i for $i \geq 1$. Then

$$n = \sum_i n_i \quad \text{and} \quad 2(n-1) = 2m = \sum_i i n_i.$$

(a) Prove that $n_1 = 2 + n_3 + 2n_4 + 3n_5 + 4n_6 + \ldots$.

(b) A tree T has three vertices of degree 2, five vertices of degree 3, two vertices of degree 4, and no vertices of degree 5 or more. According to the formula in (a), how many end-vertices does T have?

4.20 Prove or disprove:

(a) If G is a graph of order n and size m with three cycles, then $m \geq n+2$.

(b) There exist exactly two regular trees.

4.21 Let T be a tree of order n. Prove that T is isomorphic to a subgraph of $\overline{C}_{n+2}$.

4.22 Let T be a tree of order n. Show that the size of the complement $\overline{T}$ of T is the same as the size of K_{n-1}.

4.23 Find all trees T such that $\overline{T}$ is also a tree.

4.24 (a) Find all those graphs G of order $n \geq 4$ such that the subgraph induced by every three vertices of G is a tree, or show that no such graph exists.

(b) State and solve a generalization of the problem in (a).

4.3 The Minimum Spanning Tree Problem

If the seven villages illustrated in Figure 4.1 really did exist, quite possibly the villages developed one by one; and as a new village developed, a new road was constructed that connected this village to the previously developed villages. For example, suppose that the three villages v_1, v_2, and v_3 already existed with a road between v_1 and v_2 and a road between v_2 and v_3. Furthermore, suppose that the settlement v_4 developed into a village. Then it would be logical to construct a paved road between v_4 and one of v_1, v_2, and v_3. Of course, it might be preferable to construct a road between v_4 and an intermediate location along an existing road, which in turn may lead to a new development at this junction. But let's assume that this doesn't occur. However, just as with many decisions in life, the decision as to which road should be built would most likely be a financial one.

On the other hand, suppose that initially no roads existed between any pair of the villages $v_1, v_2, \ldots, v_7$ (as might be the case if these are the seven Olympic dormitories that are to be constructed to house the athletes at a forthcoming Olympic Games). Then we need to construct roads between pairs of dormitories. Which roads will be constructed is quite possibly a financial decision here as well. Before proceeding further, let us consider a new concept.

If a connected graph G of order n has no cycles, then, of course, G is a tree. On the other hand, suppose that G contains cycles. Let e_1 be an edge lying on a cycle of G. By Theorem 4.1, e_1 is not a bridge and $G - e_1$ is connected. If $G - e_1$ contains cycles, then let e_2 be an edge lying on a cycle of $G - e_1$. Then $G - e_1 - e_2$ is connected. Eventually, we arrive at a set $X = \{e_1, e_2, \ldots, e_k\}$ of edges of G such that $G - X$ is a tree. The tree $G - X$ just constructed is a subgraph of G that has the same vertex set as G.

We can look at the observation we just made in another way. Let G be a connected graph. Consider the empty graph H with vertex set $V(G)$. Add an edge f_1 of G to H. Then add another edge f_2 of G to H. Next, add another edge f_3 of G to H, where $f_3 \notin \{f_1, f_2\}$ and such that no cycle is produced. We continue this until we have added edges $f_1, f_2, \ldots, f_{n-1}$ of G to H, producing a graph F of order n, size $n - 1$, and no cycles. By Theorem 4.8, F is a tree with $V(F) = V(G)$. We know it's possible to construct a tree in this manner as we can always choose the edges of $G - X$ mentioned above.

We have just described two ways of producing trees T that are subgraphs of a given connected graph G such that $V(T) = V(G)$. Recall that a subgraph H of a graph G is a **spanning subgraph** of G if H contains every vertex of G. A spanning subgraph H of a connected graph G such that H is a tree is called a **spanning tree** of G. For the connected graph G of Figure 4.7, two different spanning trees T_1 and T_2 of G are also shown in Figure 4.7. We have now observed the following.

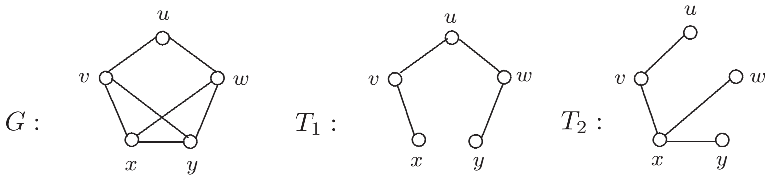


Figure 4.7: Two spanning trees in a graph

Theorem 4.10 *Every connected graph contains a spanning tree.*

Once again, let's return to the example we considered in Figure 4.1, where there are seven villages and six roads. The graph describing this situation is a tree. Hence it is possible to travel between every two villages. Indeed, there is a unique path between every two villages. To travel between villages v_1 and v_6, we are forced to pass through v_2, v_3, and v_4, even if we didn't want to. Therefore, the trip between v_1 and v_6 may be inconvenient. Of course, to make the trip between pairs of the villages more convenient, we could always build a new road (between v_1 and v_6, say). However, this would cost more money (possibly a great deal of money). But how was it decided initially that the six roads in Figure 4.1 were the ones to be constructed? Certainly, whichever roads were chosen should produce a connected graph. If the resulting graph contains a cycle, then there are edges in the graph that are not bridges. That is, if producing a connected graph was our primary goal, then we could have accomplished this for less money by constructing roads so that the resulting graph is a tree. But how did we choose those *particular* six roads to build?

Suppose that we have a number of villages (such as the villages $v_1, v_2, \ldots, v_7$) and we would like to build roads as cheaply as possible so that, at the conclusion, the resulting graph is connected. How do we do this? Assume that we have an accurate estimate of the cost of building a road between each pair of villages. If the cost of building a road between some pair of villages is exorbitant (because any such road would have to pass through quicksand, private property, or through or over a mountain, for example), then we do not even consider building such a road. This problem can be stated in terms of graphs.

Let G be a connected graph each of whose edges is assigned a number (called the **cost** or **weight** of the edge). We denote the weight of an edge e of G by $w(e)$. Recall that such a graph is called a **weighted graph**. For each subgraph H of G, the **weight** $w(H)$ of H is defined as the sum of the weights of its edges, that is,

$$w(H) = \sum_{e \in E(H)} w(e).$$

We seek a spanning tree of G whose weight is minimum among all spanning trees of G. Such a spanning tree is called a **minimum spanning tree**. The problem of finding a minimum spanning tree in a connected weighted graph is called the **Minimum Spanning Tree Problem**.

The importance of the Minimum Spanning Tree Problem is due to its applications in the design of computer, communications, and transportation networks. The history of this problem was researched by Ronald L. Graham and Pavol Hell in 1985. (We will encounter Graham again in Chapter 11.) They concluded that the Minimum Spanning Tree Problem was initially formulated by Otakar Borůvka in 1926 because of his interest in the most economical layout of a power-line network. He also gave the first solution of the problem. Prior to Borůvka, however, the anthropologist Jan Czekanowski's work on classification schemes led him to consider ideas closely related to the Minimum Spanning Tree Problem.

Over the years, this problem has been solved in a variety of ways using a number of algorithms. One of the best known was discovered by John Bernard Kruskal (born in 1928). Kruskal is from a family of five children, three boys and two girls. All boys became mathematicians. Kruskal received his Ph.D. from Princeton in 1954. His advisors were Paul Erdős and Roger Lyndon. Throughout his life he has been an active researcher, with much of his work in mathematics and linguistics. He spent much of his life working at Bell Laboratories. However, it was only two years after completing his doctoral degree that the paper was published containing the algorithm that bears his name.

Kruskal's Algorithm: For a connected weighted graph G, a spanning tree T of G is constructed as follows: For the first edge e_1 of T, we select any edge of G of minimum weight and for the second edge e_2 of T, we select any remaining edge of G of minimum weight. For the third edge e_3 of T, we choose any remaining edge of G of minimum weight that does not produce a cycle with the previously selected edges. We continue in this manner until a spanning tree is produced.

Figure 4.8 shows how a spanning tree of a connected weighted graph is constructed using Kruskal's Algorithm. We now show that Kruskal's Algorithm produces a minimum spanning tree in every connected weighted graph.

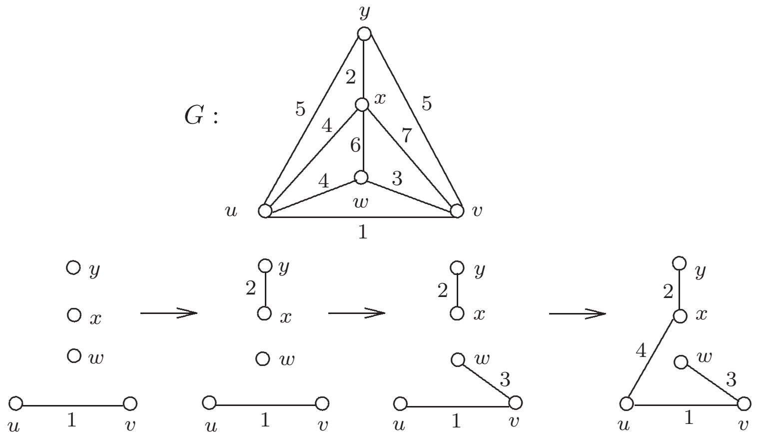


Figure 4.8: Constructing a spanning tree by Kruskal's Algorithm

Theorem 4.11 *Kruskal's Algorithm produces a minimum spanning tree in a connected weighted graph.*

Proof. [proof by contradiction] Let G be a connected weighted graph of order n and let T be a spanning tree obtained by Kruskal's Algorithm, where the edges of T are selected in the order e_1, e_2, ..., e_{n-1}. Necessarily then, $w(e_1) \leq w(e_2) \leq \ldots \leq w(e_{n-1})$ and the weight of T is

$$w(T) = \sum_{i=1}^{n-1} w(e_i).$$

We show that T is a minimum spanning tree of G. Assume, to the contrary, that T is not a minimum spanning tree. Among all minimum spanning trees of G, let H be one that has a maximum number of edges in common with T. Since H and T are not identical, there is at least one edge of T that is not in H. Let e_i be the first edge of T that is not in H. Therefore, if $i > 1$, then the edges e_1, e_2, ..., e_{i-1} belong to both H and T. Now define $G_0 = H + e_i$. Then G_0 has a cycle C. Since T has no cycle, there is an edge e_0 on C that is not in T. The graph $T_0 = G_0 - e_0$ is therefore a spanning tree of G and

$$w(T_0) = w(H) + w(e_i) - w(e_0).$$

Since H is a minimum spanning tree of G, it follows that $w(H) \leq w(T_0)$. Consequently, $w(H) \leq w(H) + w(e_i) - w(e_0)$ and so $w(e_0) \leq w(e_i)$. By Kruskal's Algorithm, certainly $w(e_0) = w(e_i)$ if $i = 1$. Suppose then that $i > 1$. By Kruskal's Algorithm, e_i is an edge of minimum weight that can be added to the edges $e_1, e_2, \ldots, e_{i-1}$ without producing a cycle. However, e_0 can also be added to $e_1, e_2, \ldots, e_{i-1}$ without producing a cycle. Thus $w(e_i) \leq w(e_0)$, which implies that $w(e_i) = w(e_0)$ when $i > 1$ as well. Therefore, $w(T_0) = w(H)$ and so T_0 is also a minimum spanning tree of G. However, T_0 has more edges in common with T than H does, which is a contradiction. ∎

Another well-known algorithm for finding a minimum spanning tree in a connected weighted graph was developed by Robert Clay Prim (born in 1921). Like Kruskal, he received his Ph.D. from Princeton (in 1949). He was vice president of research at the Sandia Corporation. The paper containing the algorithm that bears his name was published in 1957.

Prim's Algorithm: For a connected weighted graph G, a spanning tree T of G is constructed as follows: For an arbitrary vertex u for G, an edge of minimum weight incident with u is selected as the first edge e_1 of T. For subsequent edges $e_2, e_3, \cdots, e_{n-1}$, we select an edge of minimum weight among those edges having exactly one of its vertices incident with an edge already selected.

Figure 4.9 illustrates how to construct a spanning tree of a connected weighted graph by Prim's Algorithm. Again, a tree obtained by Prim's Algorithm is also a minimum spanning tree, as we show next.

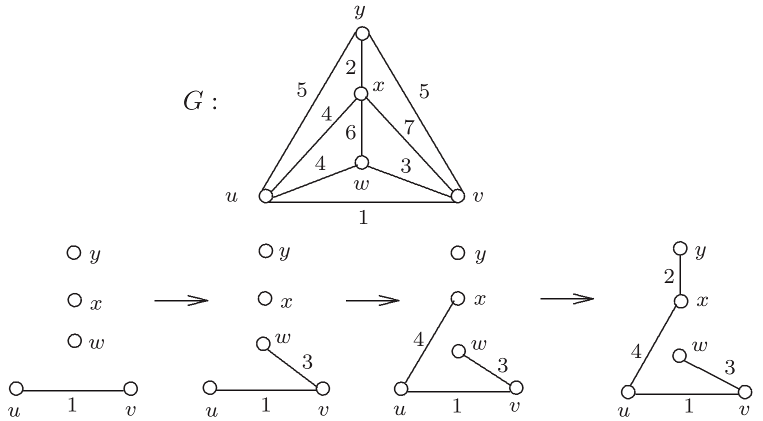


Figure 4.9: Constructing a spanning tree by Prim's Algorithm

Theorem 4.12 *Prim's Algorithm produces a minimum spanning tree in a connected weighted graph.*

Proof. [proof by contradiction] Let G be a nontrivial connected weighted graph of order n and let T be a spanning tree obtained by Prim's Algorithm, where the edges of T are selected in the order $e_1, e_2, \ldots, e_{n-1}$, and where e_1 is incident with a given vertex u. Thus the weight of T is

$$w(T) = \sum_{i=1}^{n-1} w(e_i).$$

Assume, to the contrary, that T is not a minimum spanning tree. Let $\mathcal{H}$ be the set of all minimum spanning trees of G having a maximum number of edges in common with T. By assumption, $T \notin \mathcal{H}$. If no tree in $\mathcal{H}$ contains e_1, let H be any tree in $\mathcal{H}$; otherwise, let H be a tree in $\mathcal{H}$ that contains the edges $e_1, e_2, \ldots, e_k$ for some integer k with $1 \leq k < n-1$ but for which no tree in $\mathcal{H}$ contains all of the edges $e_1, e_2, \ldots, e_{k+1}$. Hence no tree in $\mathcal{H}$ contains all of the edges $e_1, e_2, \ldots, e_{k+1}$, where $0 \leq k < n-1$. If $k = 0$, let $U = \{u\}$; while if $k \geq 1$, let U be the set of $k+1$ vertices that are incident with one or more of the edges $e_1, e_2, \ldots, e_k$. By Prim's Algorithm, e_{k+1} joins a vertex of U and a vertex of $V(T) - U$.

The subgraph $H + e_{k+1}$ therefore contains a cycle C and e_{k+1} is on C. Necessarily, C contains an edge e_0 distinct from e_{k+1} such that e_0 also joins a vertex of U and a vertex of $V(T) - U$. By the construction of T from Prim's Algorithm, $w(e_{k+1}) \leq w(e_0)$. Now $T' = H + e_{k+1} - e_0$ is a spanning tree of G whose weight is

$$w(T') = w(H) + w(e_{k+1}) - w(e_0).$$

Since H is a minimum spanning tree, $w(H) \leq w(T')$ and so $w(H) \leq w(H) + w(e_{k+1}) - w(e_0)$, which implies that $w(e_0) \leq w(e_{k+1})$. Consequently, $w(e_0) = w(e_{k+1})$ and $w(H) = w(T')$. Therefore, T' is also a minimum spanning tree of G. If e_0 does not belong to T, then T' is a minimum spanning tree having more edges in common with T than H does, which is impossible since $H \in \mathcal{H}$. Hence e_0 belongs to T, which implies that T' has the same number of edges in common with T as H does and so $T' \in \mathcal{H}$. Necessarily, $e_0 = e_j$ for some $j > k+1$. Since T' contains all of the edges $e_1, e_2, \ldots, e_{k+1}$, this contradicts the defining property of H. ■

Exercises for Section 4.3

4.25 Determine all spanning trees for the graphs G and H in Figure 4.10. Which of these spanning trees are isomorphic?

4.26 Prove that an edge e of a connected graph is a bridge if and only if e belongs to every spanning tree of G.

4.27 Apply both Kruskal's and Prim's Algorithms to find a minimum spanning tree in the weighted graph in Figure 4.11. In each case, show how this tree is constructed, as in Figures 4.8 and 4.9.

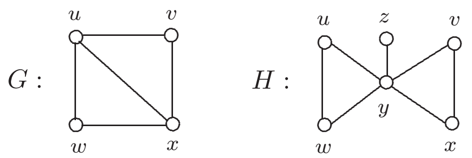


Figure 4.10: The graphs in Exercise 4.25

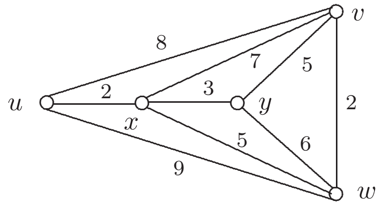


Figure 4.11: The weighted graph in Exercise 4.27

4.28 Apply both Kruskal's and Prim's Algorithms to find a minimum spanning tree in the weighted graph in Figure 4.12. In each case, show how this tree is constructed, as in Figures 4.8 and 4.9.

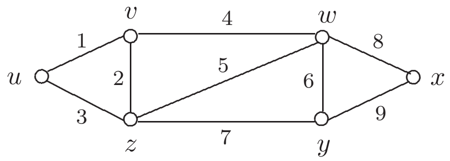


Figure 4.12: The weighted graph in Exercise 4.28

4.29 Let G be a connected weighted graph whose edges have distinct weights. Prove that G has a unique minimum spanning tree.

4.30 Let G be a connected weighted graph and T a minimum spanning tree of G. Show that T is a unique minimum spanning tree of G if and only if the weight of each edge e of G that is not in T exceeds the weight of every other edge on the cycle in $T + e$.

4.31 Show, for each integer $k \geq 2$, that there exists a connected weighted graph containing exactly k unequal minimum spanning trees.

4.4 Excursion: The Number of Spanning Trees

We have already mentioned that every connected graph G contains a spanning tree. It is essential, of course, that G is connected, for otherwise G has no spanning trees. Also, if G is itself a tree, then G contains exactly one spanning tree, namely G itself. On the other hand, if G is a connected labeled graph that is not a tree, then G contains more than one spanning tree. But how many? In this section, we are concerned with counting the number of (unequal) spanning trees of a labeled connected graph. To simplify the consideration of a number of graphs, we intend that vertex labels are present even if none are shown. Spanning trees with different edge sets are therefore different. We now consider two examples.

Example 4.13 *Determine the number of spanning trees of the graph G of Figure* 4.13.

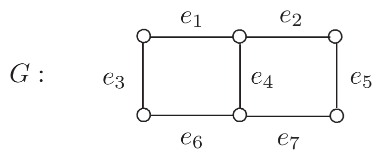


Figure 4.13: The graph in Example 4.13

Solution. Observe that at least one edge of each cycle of G must be absent from every spanning tree of G. We consider the number of spanning trees of G that (1) do not contain e_4 and (2) contain e_4. First, any spanning tree that does not contain e_4 must contain exactly five of the six edges e_1, e_2, e_3, e_5, e_6, e_7. Hence there are six spanning trees of G that do not contain e_4. Second, any spanning tree that contains e_4 must not contain exactly one of e_1, e_3, e_6 and must not contain exactly one of e_2, e_5, e_7. Therefore, there are $3 \cdot 3 = 9$ spanning trees that contain e_4. So there are $6 + 9 = 15$ spanning trees of G. $\diamondsuit$

Example 4.14 *Determine the number of spanning trees of the graph G of Figure* 4.14.

Solution. At least one of the edges e_3, e_5, e_6 is absent in each spanning tree of G. This divides the spanning trees of G into three mutual disjoint categories.

Category 1. *Those spanning trees containing none of the edges e_3, e_5, e_6.* Then a spanning tree is obtained by deleting any of the remaining six edges e_1, e_2, e_4, e_7, e_8, e_9. Hence there are 6 such spanning trees.

Category 2. *Those spanning trees containing exactly one of the edges e_3, e_5, e_6.* Suppose first that e_3 belongs to a spanning tree but e_5 and e_6 do not. Then

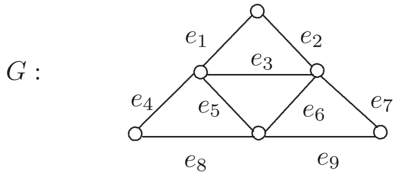


Figure 4.14: The graph in Example 4.14

either e_1 or e_2 does not belong to a spanning tree and exactly one of e_4, e_7, e_8, e_9 does not belong to a spanning tree. Therefore, the number of spanning trees of G containing e_3 but neither e_5 nor e_6 is $2 \cdot 4 = 8$. By the symmetry of G, the number of spanning trees of G containing exactly one of the edges e_3, e_5, e_6 is $3 \cdot 8 = 24$.

Category 3. *Those spanning trees containing exactly two of the edges* e_3, e_5, e_6. Suppose first that e_3 and e_5 belong to a spanning tree but e_6 does not. Then such a spanning tree is obtained by deleting exactly one edge in each of the following three pairs of edges: (1) e_1, e_2; (2) e_4, e_8; (3) e_7, e_9. Therefore, the number of spanning trees of G containing e_3 and e_5 but not e_6 is $2^3 = 8$. Again, by symmetry, the number of spanning trees of G containing exactly two of the three edges e_3, e_5, e_6 is $3 \cdot 8 = 24$.

Therefore, the total number of spanning trees of G is $6 + 24 + 24 = 54$. $\diamond$

We now turn to the problem of determining the number of spanning trees in a complete graph K_n. Since K_n is a tree for $n = 1$ or $n = 2$, we need only consider $n \geq 3$. Since K_3 is a cycle of length 3, each spanning tree is obtained by deleting one of the three edges, that is, the number of spanning trees of K_3 is 3 (see Figure 4.15).

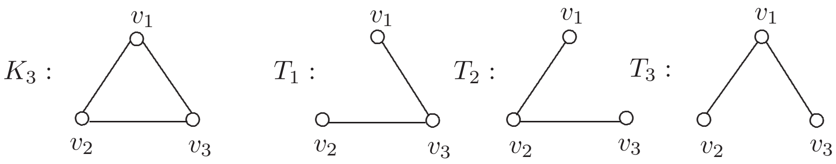


Figure 4.15: The spanning trees of K_3

Determining the number of spanning trees of K_4 (see Figure 4.16) is more troublesome, but one way of computing this is by observing that any spanning tree of K_4 contains (1) none, (2) exactly one, or (3) exactly two of the edges v_1v_2, v_1v_3, and v_2v_3. Since the number of spanning trees in each of these three cases is 1, 6, and 9, respectively, the number of spanning trees of K_4 is $1+6+9 = 16$.

Actually, computing the total number of of spanning trees of the graph $G \cong K_n$, where $V(G) = \{v_1, v_2, \ldots, v_n\}$, is the same as computing the number

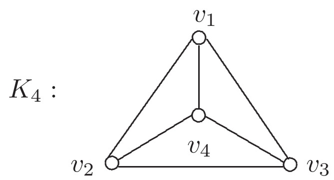


Figure 4.16: The complete graph K_4

of distinct trees with vertex set $\{v_1, v_2, \ldots, v_n\}$. The following formula was established in 1889 by Arthur Cayley and is often referred to as the **Cayley Tree Formula**. As a consequence of this formula, it becomes clear why there are 16 spanning trees of K_4.

Theorem 4.15 *The number of distinct trees of order n with a specified vertex set is n^{n-2}.*

We have encountered Cayley several times already. Arthur Cayley was born on August 16, 1821 in Richmond, Surrey, England. He showed great skill with numerical calculations as a youngster. In 1838 he entered Trinity College, Cambridge and had three papers published while still an undergraduate. He graduated in 1842 and then spent four years teaching as a Cambridge fellow, at which time he continued to publish at a high rate. When his fellowship expired, Cayley found himself without a position. He then studied law and became a lawyer in 1849, an occupation he continued for the next 14 years. Although a skilled lawyer, Cayley considered this as a way to make money so that he could do what he really enjoyed: mathematics. During his 14 years as a lawyer, Cayley authored more than 200 mathematical papers.

In 1849 Cayley wrote a paper on permutations connecting his ideas with those of Augustin Louis Cauchy. In 1854 he wrote two remarkable papers on abstract groups. At that time the only known groups were permutation groups. Cayley defined an abstract group and gave a table to display the multiplication in the group. He also recognized that matrices formed groups.

In 1863 Cayley was appointed a professor of Pure Mathematics at Cambridge. Although Cayley was earning only a fraction of what he earned as a lawyer, he was happier as he was able to work on mathematics full-time. He was a prolific researcher his entire life and when he died on January 26, 1895 in Cambridge, he had published over 900 papers, a number exceeded only by Leonhard Euler, Paul Erdős, and Cauchy.

Although we will not be presenting a proof of Theorem 4.15, there are several quite different proofs of this formula. In fact, John W. Moon wrote a paper in 1967 in which he describes ten different proofs of this famous theorem (and even more proofs have been found since). Moon, a professor at the University of Alberta in Canada throughout much of his academic career, did a great deal of research on trees (and on a class of digraphs that we will visit in Chapter 7). With an interest in music, as has been the case with a number of mathematicians, Moon has a philosophy about the creative aspect of mathematics that is no doubt not unlike that of many mathematicians:

> *The sense of pleasure and satisfaction that comes when one has discovered something new is hard to describe to non-mathematicians; and sometimes there is almost a sense of awe, that one has been privileged to have a peek at what lies behind the mystery of things. And when you can share the experience with a co-worker, so much the better.*

There is a method of determining the number of spanning trees of any graph. The next theorem is implicit in the work of Gustav Kirchhoff who was born in Königberg, Prussia in 1824. (We will visit that city again in Chapter 6.) Kirchhoff is well-known for his research on electrical currents, which he announced in 1845. This led to Kirchhoff's laws, the first of which states that the sum of the currents into a vertex equals the sum of the currents out of the vertex. Two years later, in 1847, he graduated from the University of Königberg. It was during that year that he published the paper that led to his theorem on counting spanning trees. Kirchhoff spent much of his life working on experimental physics. When his health began to fail, he accepted a position as chair of mathematical physics in 1875 (12 years before he died) in Berlin as this did not present the problems his poor health was causing him in carrying out experiments. Since the proof of the theorem on counting spanning trees is complex, we will not include it.

By a **cofactor** of an $n \times n$ matrix $M = [m_{ij}]$, we mean $(-1)^{i+j}\det(M_{ij})$, where $\det(M_{ij})$ indicates the determinant of the $(n-1)\times(n-1)$ submatrix M_{ij} of M, obtained by deleting row i and column j of M. The following result is often called the **Matrix Tree Theorem**.

Theorem 4.16 *Let G be a graph with $V(G) = \{v_1, v_2, \ldots, v_n\}$, let $A = [a_{ij}]$ be the adjacency matrix of G, and let $C = [c_{ij}]$ be the $n \times n$ matrix, where*

$$c_{ij} = \begin{cases} \deg v_i & \text{if } i = j \\ -a_{ij} & \text{if } i \neq j. \end{cases}$$

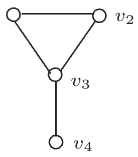
Then the number of spanning trees of G is the value of any cofactor of C.

We illustrate the Matrix Tree Theorem with a simple example. The graph G of Figure 4.17 clearly contains three spanning trees. The adjacency matrix A and the matrix C are also shown in Figure 4.17.

The $(3,3)$-cofactor of C is expanded along the 3rd row, obtaining

$$\begin{vmatrix} 2 & -1 & 0 \\ -1 & 2 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 0\begin{vmatrix} -1 & 0 \\ 2 & 0 \end{vmatrix} - 0\begin{vmatrix} 2 & 0 \\ -1 & 0 \end{vmatrix} + 1\begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix}$$
$$= 0 \cdot 0 - 0 \cdot 0 + 1 \cdot 3 = 3,$$

as expected.

$G:$ 

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \qquad C = \begin{bmatrix} 2 & -1 & -1 & 0 \\ -1 & 2 & -1 & 0 \\ -1 & -1 & 3 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix}$$

Figure 4.17: Illustrating the Matrix Tree Theorem

Exercises for Section 4.4

4.32 Show that there is only one positive integer k such that no graph contains exactly k spanning trees.

4.33 Let F be a subgraph of a connected graph G. Prove that F is a subgraph of some spanning tree of G if and only if F contains no cycles.

4.34 (a) Find the number of spanning trees in the graph G of Figure 4.18.

(b) Find the number of spanning trees in the graph G_k for $k \geq 5$ of Figure 4.18. [Note that (a) is the case where $k = 4$.]

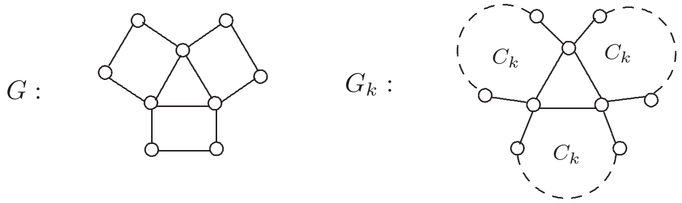


Figure 4.18: The graphs in Exercise 4.34

4.35 (a) Find the number of spanning trees in the graph G of Figure 4.19.

(b) Find the number of spanning trees in the graph G_k for $k \geq 6$ of Figure 4.19. [Note that (a) is the case where $k = 5$.]

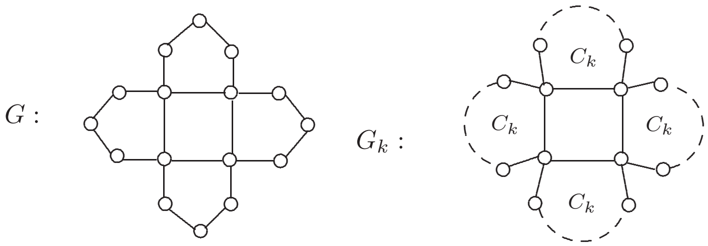


Figure 4.19: The graphs in Exercise 4.35

4.36 Find the number of spanning trees in the graph G of Figure 4.20.

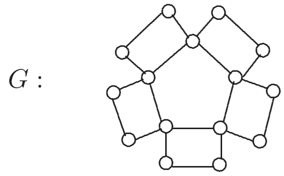


Figure 4.20: The graph in Exercise 4.36

4.37 (a) According to the Cayley Tree Formula, how many distinct trees are there with vertex set $S = \{u, v, w, x, y\}$?

(b) Divide the trees in (a) into classes so that two trees are in the same class if and only if they are isomorphic. Determine the number of such classes and the number of trees in each class by considering the number of ways each can be labeled with the elements of S.

4.38 Use the Matrix Tree Theorem to confirm the number of distinct trees with vertex set $\{v_1, v_2, v_3, v_4\}$.

4.39 (a) Use the Matrix Tree Theorem to determine the number of spanning trees of the graph of Figure 4.21.

(b) Draw all spanning trees in the graph G of Figure 4.21.

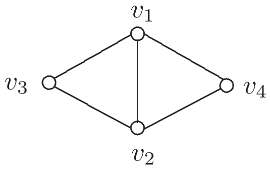


Figure 4.21: The graph in Exercise 4.39

4.40 Let T and T' be two spanning trees of a connected graph G of order n. Show that there exists a sequence $T = T_0, T_1, \ldots, T_k = T'$ of spanning trees of G such that T_i and T_{i+1} have $n-2$ edges in common for each i with $1 \leq i \leq k-1$.